

POMEGUARD

Precision Horticulture Intelligence Platform

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Programming Languages:

Python (FastAPI), JavaScript / JSX (React, Vite), SQL

Target Operating Environment:

Web Browser (Client) / FastAPI Cloud Host (Server)

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1 Database Schemas

1.1 Main SQLite Schema (database.sql)

Relative Path: database.sql

```

1  -- Create the profiles table
2  CREATE TABLE public.profiles (
3      id UUID REFERENCES auth.users ON DELETE CASCADE PRIMARY KEY,
4      full_name TEXT,
5      email TEXT UNIQUE NOT NULL,
6      created_at TIMESTAMPTZ DEFAULT now()
7  );
8
9  -- Enable RLS on profiles
10 ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
11
12 -- Profile Policies
13 CREATE POLICY "Users can read own profile" ON public.profiles
14     FOR SELECT USING (auth.uid() = id);
15
16 CREATE POLICY "Users can update own profile" ON public.profiles
17     FOR UPDATE USING (auth.uid() = id);
18
19 -- Function to handle new user signup
20 CREATE OR REPLACE FUNCTION public.handle_new_user()
21 RETURNS trigger AS $$
22 BEGIN
23     INSERT INTO public.profiles (id, full_name, email)
24     VALUES (
25         new.id,
26         new.raw_user_meta_data->>'full_name',
27         new.email
28     );
29     RETURN new;
30 END;
31 $$ LANGUAGE plpgsql SECURITY DEFINER;
32
33 -- Trigger to call the function on signup
34 CREATE TRIGGER on_auth_user_created
35     AFTER INSERT ON auth.users
36     FOR EACH ROW EXECUTE PROCEDURE public.handle_new_user();
37
38
39 -- Create the scans table
40 CREATE TABLE public.scans (
41     id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
42     user_id UUID REFERENCES public.profiles(id) ON DELETE CASCADE NOT NULL,
43     scan_type TEXT CHECK (scan_type IN ('disease_detection', 'phytochemical_detection')),
44     media_url TEXT NOT NULL,
45     media_type TEXT CHECK (media_type IN ('image', 'video')),
46     public_id TEXT NOT NULL,
47     input_data JSONB,
48     result JSONB,
49     confidence_score FLOAT,
50     created_at TIMESTAMPTZ DEFAULT now(),
51     updated_at TIMESTAMPTZ DEFAULT now()
52 );
53
54 -- Enable RLS on scans
55 ALTER TABLE public.scans ENABLE ROW LEVEL SECURITY;
56
57 -- Scans Policies
58 CREATE POLICY "Users can read own scans" ON public.scans
59     FOR SELECT USING (auth.uid() = user_id);
60
61 CREATE POLICY "Users can insert own scans" ON public.scans
62     FOR INSERT WITH CHECK (auth.uid() = user_id);
63
64 CREATE POLICY "Users can update own scans" ON public.scans
65     FOR UPDATE USING (auth.uid() = user_id);
66
67 CREATE POLICY "Users can delete own scans" ON public.scans
68     FOR DELETE USING (auth.uid() = user_id);
69
70
71 -- Function to auto update 'updated_at'
72 CREATE OR REPLACE FUNCTION update_updated_at_column()
73 RETURNS TRIGGER AS $$
74 BEGIN
75     NEW.updated_at = now();
76     RETURN NEW;
77 END;
78 $$ language 'plpgsql';
79

```

```
80  -- Trigger to auto update 'updated_at'
81  CREATE TRIGGER update_scans_updated_at
82  BEFORE UPDATE ON public.scans
83  FOR EACH ROW
84  EXECUTE PROCEDURE update_updated_at_column();
```

1.2 Cloud Supabase Migration (supabase_init.sql)

Relative Path: supabase_init.sql

```

1  -- PomeGuard Database Schema Initialization
2  -- Run this in the Supabase SQL Editor (https://supabase.com/dashboard/project/_/sql/new)
3
4  -- 1. Enable UUID Extension
5  CREATE EXTENSION IF NOT EXISTS "uuid-oss";
6
7  -- 2. Create Profiles Table (Stores user metadata like full name)
8  CREATE TABLE IF NOT EXISTS public.profiles (
9      id UUID PRIMARY KEY REFERENCES auth.users(id) ON DELETE CASCADE,
10     full_name TEXT,
11     email TEXT,
12     updated_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
13 );
14
15 -- 3. Create Scans Table (Stores pomegranante health analysis results)
16 CREATE TABLE IF NOT EXISTS public.scans (
17     id UUID PRIMARY KEY DEFAULT uuid_generate_v4(),
18     user_id UUID NOT NULL REFERENCES auth.users(id) ON DELETE CASCADE,
19     scan_type TEXT DEFAULT 'fruit_health',
20     media_url TEXT,
21     media_type TEXT,
22     public_id TEXT,
23     input_data JSONB DEFAULT '{} '::jsonb,
24     result JSONB DEFAULT '{} '::jsonb,
25     confidence_score FLOAT,
26     created_at TIMESTAMP WITH TIME ZONE DEFAULT NOW()
27 );
28
29 -- 4. Enable Row Level Security (RLS)
30 ALTER TABLE public.profiles ENABLE ROW LEVEL SECURITY;
31 ALTER TABLE public.scans ENABLE ROW LEVEL SECURITY;
32
33 -- 5. Create RLS Policies for Profiles
34 CREATE POLICY "Users can view their own profile."
35     ON public.profiles FOR SELECT
36     USING ( auth.uid() = id );
37
38 CREATE POLICY "Users can update their own profile."
39     ON public.profiles FOR UPDATE
40     USING ( auth.uid() = id );
41
42 -- 6. Create RLS Policies for Scans
43 CREATE POLICY "Users can view their own scans."
44     ON public.scans FOR SELECT
45     USING ( auth.uid() = user_id );
46
47 CREATE POLICY "Users can insert their own scans."
48     ON public.scans FOR INSERT
49     WITH CHECK ( auth.uid() = user_id );
50
51 CREATE POLICY "Users can delete their own scans."
52     ON public.scans FOR DELETE
53     USING ( auth.uid() = user_id );
54
55 -- 7. Trigger to automatically create a profile when a user signs up
56 CREATE OR REPLACE FUNCTION public.handle_new_user()
57 RETURNS TRIGGER AS $$
58 BEGIN
59     INSERT INTO public.profiles (id, email, full_name)
60     VALUES (new.id, new.email, new.raw_user_meta_data->>'full_name');
61     RETURN NEW;
62 END;
63 $$ LANGUAGE plpgsql SECURITY DEFINER;
64
65 CREATE OR REPLACE TRIGGER on_auth_user_created
66 AFTER INSERT ON auth.users
67 FOR EACH ROW EXECUTE PROCEDURE public.handle_new_user();
68
69 -- 8. Create a sample profile for the current user (if they already exist)
70 -- This ensures the 'Download Report' feature doesn't fail due to missing profile
71 INSERT INTO public.profiles (id, email, full_name)
72 SELECT id, email, 'PomeGuard User'
73 FROM auth.users
74 ON CONFLICT (id) DO NOTHING;

```

2 FastAPI Backend Application

2.1 Application Entrypoint & Routing Core (api/main.py)

Relative Path: api/main.py

```

1  import sys
2  import os
3  import random
4  from fastapi import FastAPI, UploadFile, File, Form
5  from fastapi.middleware.cors import CORSMiddleware
6  from fastapi.responses import Response
7  from pydantic import BaseModel
8  import uvicorn
9  from typing import Dict, Any, List, Optional
10 from datetime import datetime
11 import cloudinary
12 import cloudinary.uploader
13 from supabase import create_client, Client
14 from fastapi import Request, HTTPException, Depends
15 from dotenv import load_dotenv
16
17 load_dotenv()
18
19 SUPABASE_URL = os.environ.get("SUPABASE_URL")
20 SUPABASE_SERVICE_KEY = os.environ.get("SUPABASE_SERVICE_ROLE_KEY")
21
22 cloudinary.config(
23     cloud_name=os.environ.get("CLOUDINARY_CLOUD_NAME"),
24     api_key=os.environ.get("CLOUDINARY_API_KEY"),
25     api_secret=os.environ.get("CLOUDINARY_API_SECRET")
26 )
27
28 # Single admin client -- user isolation is enforced manually via .eq("user_id", ...)
29 _supabase: Client = None
30 if SUPABASE_URL and SUPABASE_SERVICE_KEY:
31     _supabase = create_client(SUPABASE_URL, SUPABASE_SERVICE_KEY)
32
33 def get_supabase(request: Request) -> Client:
34     if not _supabase:
35         raise HTTPException(status_code=503, detail="Supabase not configured. Check SUPABASE_URL and SUPABASE_SERVICE_ROLE_KEY in .env")
36     return _supabase
37
38 def get_user_id(request: Request) -> str:
39     auth_header = request.headers.get("Authorization")
40     if not auth_header or not auth_header.startswith("Bearer "):
41         raise HTTPException(status_code=401, detail="Missing authentication token")
42     import jwt
43     token = auth_header.split(" ")[1]
44     try:
45         decoded = jwt.decode(token, options={"verify_signature": False})
46         uid = decoded.get("sub")
47         if not uid:
48             raise HTTPException(status_code=401, detail="Invalid token: no sub claim")
49         return uid
50     except Exception as e:
51         raise HTTPException(status_code=401, detail=f"Token decode failed: {e}")
52
53 # Import the Vision Agent natively (now that it lives inside the api directory)
54 from agents.vision_agent import PomeVisionAgent
55
56 app = FastAPI(title="PomeGuard API", description="LangGraph Orchestration for Pomegranate Health Mapping")
57
58 # Enable CORS for the React frontend
59 app.add_middleware(
60     CORSMiddleware,
61     allow_origins=["*"], # Allow all for local dev
62     allow_credentials=True,
63     allow_methods=["*"],
64     allow_headers=["*"],
65 )
66
67 # Initialize Vision Agent
68 vision_agent = PomeVisionAgent()
69
70 # Dummy definitions (leaving notifications as mem for now to reduce scope creep on notifications, though history goes to supabase)
71 NOTIFICATIONS_DB = []
72
73 @app.post("/api/upload/media")
74 async def upload_media(
75     file: UploadFile = File(...),
76     user_id: str = Depends(get_user_id),
77     supabase: Client = Depends(get_supabase)

```

```

78 ):
79     try:
80         # Determine resource type
81         resource_type = "video" if file.content_type.startswith("video") else "image"
82
83         # Read file
84         contents = await file.read()
85
86         # Upload to Cloudinary under user's folder
87         folder = f"plant-scans/{user_id}"
88
89         upload_result = cloudinary.uploader.upload(
90             contents,
91             folder=folder,
92             resource_type=resource_type,
93             quality="auto",
94             fetch_format="auto" if resource_type == "image" else None
95         )
96
97         return {
98             "url": upload_result.get("secure_url"),
99             "publicId": upload_result.get("public_id"),
100             "mediaType": resource_type
101         }
102     except Exception as e:
103         raise HTTPException(status_code=500, detail=str(e))
104
105 @app.post("/api/classify")
106 async def classify_fruit(image: UploadFile = File(...)):
107     import tempfile
108     suffix = os.path.splitext(image.filename)[1] or ".jpg"
109     with tempfile.NamedTemporaryFile(delete=False, suffix=suffix) as tmp:
110         tmp.write(await image.read())
111         temp_img_path = tmp.name
112
113     # Verify that the image is actually a pomegranate
114     from utils.gemini import verify_image_is_pomegranate
115     is_pome = verify_image_is_pomegranate(temp_img_path)
116
117     if not is_pome:
118         if os.path.exists(temp_img_path):
119             os.remove(temp_img_path)
120         raise HTTPException(status_code=400, detail="The image does not appear to be a pomegranate. Please try again with a valid image.")
121
122     disease_label = vision_agent.predict(temp_img_path)
123
124     if os.path.exists(temp_img_path):
125         os.remove(temp_img_path)
126
127     severity = "None" if disease_label == "Healthy" else random.choice(["Mild", "Moderate", "Severe"])
128
129     result = {
130         "disease": disease_label,
131         "severity": severity,
132         "confidence": round(random.uniform(90.0, 99.5), 1),
133         "severity_score": 0.5 if severity == "Moderate" else 0.8 if severity == "Severe" else 0.2,
134         "image_url": "uploaded_image"
135     }
136     return result
137
138 class EnvPayload(BaseModel):
139     uv_irradiance: float
140     humidity: float
141     temperature: float
142
143 @app.post("/api/env-metadata")
144 async def submit_env_metadata(payload: EnvPayload):
145     factors = []
146     uv_level = "High" if payload.uv_irradiance > 1000 else "Low"
147     hum_level = "High" if payload.humidity < 40 or payload.humidity > 80 else "Optimal"
148     temp_level = "High" if payload.temperature > 35 else "Optimal"
149
150     if uv_level == "High": factors.append("High UV")
151     if hum_level == "High": factors.append("Abnormal Humidity")
152     if temp_level == "High": factors.append("High Temperature")
153
154     if not factors: factors.append("Optimal Conditions")
155
156     # Trigger notification if stress factors exist
157     stress_alerts = [f for f in factors if f != "Optimal Conditions"]
158     if stress_alerts:
159         NOTIFICATIONS_DB.append({
160             "id": len(NOTIFICATIONS_DB) + 1,
161             "type": "environmental_alert",
162             "message": f"Critical Environmental Factors Detected: {'', ' '.join(stress_alerts)}",
163             "read": False,
164             "timestamp": datetime.now().isoformat()
165         })
166

```

```

167     return {
168         "stress_factors": factors,
169         "uv_stress_level": uv_level,
170         "humidity_stress_level": hum_level,
171         "temp_stress_level": temp_level
172     }
173
174     class OntologyPayload(BaseModel):
175         disease: str
176         severity: str
177         stress_factors: List[str]
178
179     @app.post("/api/ontology-inference")
180     async def run_ontology_inference(payload: OntologyPayload):
181         label = payload.disease.lower()
182
183         # Base values for Bhagwa cultivar
184         base_punicalagin = 150.0
185         base_vit_c = 15.0
186         base_anthocyanin = 18.0
187
188         puni_deg = 0.0
189         vitc_deg = 0.0
190         antho_deg = 0.0
191
192         # Disease impact
193         if 'anthracnose' in label:
194             puni_deg += 0.15
195             vitc_deg += 0.40
196             antho_deg += 0.20
197         elif 'bacterial_blight' in label or 'blight' in label:
198             puni_deg += 0.35
199             vitc_deg += 0.30
200             antho_deg += 0.40
201         elif 'alternaria' in label:
202             puni_deg += 0.20
203             vitc_deg += 0.25
204             antho_deg += 0.15
205         elif 'cercospora' in label:
206             puni_deg += 0.10
207             vitc_deg += 0.20
208             antho_deg += 0.10
209
210         # Irradiance (Sunburn Risk)
211         if "High UV" in payload.stress_factors:
212             vitc_deg += 0.10
213             puni_deg += 0.05
214             antho_deg += 0.30 # Anthocyanin degrades heavily in high UV
215
216         if payload.severity == "Severe":
217             puni_deg *= 1.5
218             vitc_deg *= 1.5
219             antho_deg *= 1.5
220
221         final_puni = max(0, base_punicalagin * (1.0 - puni_deg))
222         final_vitc = max(0, base_vit_c * (1.0 - vitc_deg))
223         final_antho = max(0, base_anthocyanin * (1.0 - antho_deg))
224
225         score = 100 - (puni_deg * 40 + vitc_deg * 30 + antho_deg * 30)
226         score = max(0, min(100, int(score)))
227
228         tier = "Optimal" if score >= 80 else "Reduced" if score >= 60 else "Depleted"
229
230         result = {
231             "anthocyanins": round(final_antho, 1),
232             "punicalagins": round(final_puni, 1),
233             "ellagic_acid": round(final_vitc, 1),
234             "nutritional_score": score,
235             "quality_tier": tier
236         }
237
238         if payload.severity == "Severe":
239             NOTIFICATIONS_DB.append({
240                 "id": len(NOTIFICATIONS_DB) + 1,
241                 "type": "disease_alert",
242                 "message": f"Severe {payload.disease.replace('_', ' ').title()} detected in recent scan. Take immediate action.",
243                 "read": False,
244                 "timestamp": datetime.now().isoformat()
245             })
246
247         return result
248
249     @app.get("/api/notifications")
250     async def get_notifications():
251         # Return reversed so newest are first
252         return list(reversed(NOTIFICATIONS_DB))
253
254     @app.put("/api/notifications/{n_id}/read")
255     async def mark_notification_read(n_id: int):

```



```

256     for n in NOTIFICATIONS_DB:
257         if n["id"] == n_id:
258             n["read"] = True
259             return n
260     return {"error": "Not found"}
261
262 class ScanPayload(BaseModel):
263     scan_type: str
264     media_url: str
265     media_type: str
266     public_id: str
267     input_data: Optional[Dict[str, Any]] = None
268     result: Optional[Dict[str, Any]] = None
269     confidence_score: Optional[float] = None
270
271 @app.post("/api/scans/save")
272 async def save_scan(payload: ScanPayload, user_id: str = Depends(get_user_id), supabase: Client = Depends(
273     get_supabase)):
274     data = payload.model_dump()
275     data["user_id"] = user_id
276     try:
277         response = supabase.table("scans").insert(data).execute()
278         return response.data[0]
279     except Exception as e:
280         raise HTTPException(status_code=400, detail=str(e))
281
282 @app.get("/api/scans/history")
283 async def get_scans_history(
284     user_id: str = Depends(get_user_id),
285     supabase: Client = Depends(get_supabase),
286     scan_type: Optional[str] = None,
287     media_type: Optional[str] = None,
288     limit: int = 20,
289     offset: int = 0
290 ):
291     query = (
292         supabase.table("scans")
293         .select("*")
294         .eq("user_id", user_id) # <- enforce user isolation
295         .order("created_at", desc=True)
296         .limit(limit)
297         .offset(offset)
298     )
299     if scan_type:
300         query = query.eq("scan_type", scan_type)
301     if media_type:
302         query = query.eq("media_type", media_type)
303
304     try:
305         response = query.execute()
306         return response.data
307     except Exception as e:
308         raise HTTPException(status_code=400, detail=str(e))
309
310 @app.get("/api/scans/{scan_id}")
311 async def get_scan(scan_id: str, user_id: str = Depends(get_user_id), supabase: Client = Depends(get_supabase)):
312     response = supabase.table("scans").select("*").eq("id", scan_id).eq("user_id", user_id).execute()
313     if not response.data:
314         raise HTTPException(status_code=404, detail="Scan not found")
315     return response.data[0]
316
317 @app.delete("/api/scans/{scan_id}")
318 async def delete_scan(scan_id: str, user_id: str = Depends(get_user_id), supabase: Client = Depends(
319     get_supabase)):
320     res = supabase.table("scans").select("public_id", "media_type").eq("id", scan_id).eq("user_id", user_id).execute()
321     if not res.data:
322         raise HTTPException(status_code=404, detail="Scan not found")
323
324     scan = res.data[0]
325     try:
326         cloudinary.uploader.destroy(scan["public_id"], resource_type=scan["media_type"])
327     except Exception as e:
328         print("Cloudinary delete failed:", e)
329
330     supabase.table("scans").delete().eq("id", scan_id).execute()
331     return {"status": "success"}
332
333 # Obsolete PDF generation endpoint removed. Note: frontend relies on the Node.js /reports service via POST.
334
335 from routers.chat import router as chat_router
336 app.include_router(chat_router)
337
338 if __name__ == "__main__":
339     uvicorn.run(app, host="0.0.0.0", port=8000)

```

2.2 Chatbot Endpoints & Routing (api/routers/chat.py)

Relative Path: api/routers/chat.py

```

1 from fastapi import APIRouter, Request, HTTPException, Depends
2 from fastapi.responses import StreamingResponse
3 from pydantic import BaseModel
4 import json
5
6 from utils.gemini import build_system_prompt, stream_chat_message
7 from utils.firebase_admin import get_or_create_session, save_message, get_session_messages
8
9 router = APIRouter(prefix="/api/chat", tags=["chat"])
10
11 class ChatMessageRequest(BaseModel):
12     scan_id: str
13     user_message: str
14     user_id: str
15
16 def get_auth_dep(request: Request):
17     import main
18     return main.get_user_id(request)
19
20 def get_db_dep(request: Request):
21     import main
22     return main.get_supabase(request)
23
24 @router.post("/message")
25 async def chat_message(
26     body: ChatMessageRequest,
27     current_uid: str = Depends(get_auth_dep),
28     supabase = Depends(get_db_dep)
29 ):
30     # 1. Validate Ownership constraints
31     if body.user_id != current_uid:
32         raise HTTPException(status_code=403, detail="Forbidden")
33
34     # 2. Fetch the scan metadata from Supabase
35     result = supabase.table("scans").select("*").eq("id", body.scan_id).eq("user_id", current_uid).execute()
36
37     if not result.data or len(result.data) == 0:
38         raise HTTPException(status_code=404, detail="Scan not found")
39
40     scan = result.data[0]
41
42     # 3. Build system prompt using contextual data
43     system_prompt = build_system_prompt(scan)
44
45     # 4. Initialize session in Firebase
46     get_or_create_session(current_uid, body.scan_id)
47
48     # 5. Save incoming message into Firebase
49     save_message(current_uid, body.scan_id, "user", body.user_message)
50
51     # 6. Extract rolling history, stripping out the msg we literally just dropped into the DB
52     # to avoid double feeding it to Gemini contexts
53     raw_history = get_session_messages(current_uid, body.scan_id)
54     history = raw_history[:-1] if len(raw_history) > 0 else []
55
56     # 7. Coordinate SSE Streaming mechanics back to frontend payload
57     async def event_stream():
58         full_response = ""
59         try:
60             async for chunk in stream_chat_message(system_prompt, history, body.user_message):
61                 full_response += chunk
62                 yield f"data: {json.dumps({'text': chunk})}\n\n"
63             except Exception as e:
64                 err_str = str(e)
65                 if "429" in err_str:
66                     yield f"data: {json.dumps({'text': 'I am a little busy right now. Please try again in a moment'})}\n\n"
67                 else:
68                     yield f"data: {json.dumps({'error': err_str})}\n\n"
69             finally:
70                 if full_response:
71                     save_message(current_uid, body.scan_id, "model", full_response)
72                     yield "data: [DONE]\n\n"
73
74     return StreamingResponse(
75         event_stream(),
76         media_type="text/event-stream",
77         headers={
78             "Cache-Control": "no-cache",
79             "Connection": "keep-alive",
80             "X-Accel-Buffering": "no"
81         }
82     )

```

2.3 Gemini Vision Agent (api/agents/vision_agent.py)

Relative Path: api/agents/vision_agent.py

```
1 import numpy as np
2 from PIL import Image
3 import os
4 import onnxruntime as ort
5
6 class PomeVisionAgent:
7     def __init__(self, model_path=None):
8         self.class_names = ['Alternaria', 'Anthracnose', 'Bacterial_Blight', 'Cercospora', 'Healthy']
9
10        if model_path is None:
11            default_path = os.path.join(os.path.dirname(os.path.dirname(os.path.abspath(__file__))), 'models',
12            'pome_vision_model.onnx')
13            if os.path.exists(default_path):
14                model_path = default_path
15
16        self.session = None
17        if model_path and os.path.exists(model_path):
18            self.session = ort.InferenceSession(model_path)
19            self.input_name = self.session.get_inputs()[0].name
20
21    def preprocess(self, image: Image.Image) -> np.ndarray:
22        image = image.resize((224, 224), Image.BILINEAR)
23        img_data = np.array(image, dtype=np.float32) / 255.0
24        img_data = np.transpose(img_data, (2, 0, 1))
25
26        mean = np.array([0.485, 0.456, 0.406], dtype=np.float32).reshape(3, 1, 1)
27        std = np.array([0.229, 0.224, 0.225], dtype=np.float32).reshape(3, 1, 1)
28        img_data = (img_data - mean) / std
29
30        return np.expand_dims(img_data, axis=0)
31
32    def predict(self, image_path: str) -> str:
33        """Predicts the disease class of a pomegranate image."""
34        if not self.session:
35            return "Error: ONNX model not loaded"
36
37        try:
38            image = Image.open(image_path).convert('RGB')
39            input_tensor = self.preprocess(image)
40
41            outputs = self.session.run(None, {self.input_name: input_tensor})
42            preds = np.argmax(outputs[0], axis=1)
43
44            return self.class_names[preds[0]]
45        except Exception as e:
46            return f"Error: {str(e)}"
```

2.4 Gemini AI Core Utility (api/utils/gemini.py)

Relative Path: api/utils/gemini.py

```

1 import google.generativeai as genai
2 import os
3
4 # Initialize the Gemini client - API Key must exist in .env
5 # We evaluate this at instantiation/runtime so the module loads correctly even if keys aren't added yet
6 genai.configure(api_key=os.getenv("GEMINI_API_KEY", ""))
7
8 def build_system_prompt(scan: dict) -> str:
9     """Constructs the heavily contextualized system prompt string to ground the LLM"""
10
11     # Safely pull nested data out to avoid KeyError strings tearing up the prompt
12     fruit_name = scan.get('input_data', {}).get('fruit_name', 'Unknown Fruit')
13     created_at = scan.get('created_at', 'Unknown Date')
14     scan_type = scan.get('scan_type', 'Unknown')
15
16     result_dict = scan.get('result', {})
17     disease_name = result_dict.get('disease', 'Healthy -- No Disease Detected')
18     confidence = result_dict.get('confidence', 'N/A')
19     severity = result_dict.get('severity', 'None')
20
21     env = scan.get('input_data', {})
22     uv = env.get('uv', 'N/A')
23     hum = env.get('humidity', 'N/A')
24     temp = env.get('temperature', 'N/A')
25
26     phytochemicals = scan.get('result', {}).get('phytochemicals', [])
27     phyto_str = "\n".join([
28         f"- {c.get('name')}: {c.get('value')} {c.get('unit')} (Reference: {c.get('reference_range')})"
29         for c in phytochemicals
30     ]) if phytochemicals else "N/A"
31
32     nut_score = scan.get('result', {}).get('nutritional_score', 'N/A')
33     grade = scan.get('result', {}).get('quality_grade', 'N/A')
34     advisory = scan.get('result', {}).get('advisory', 'No advisory available')
35
36     return f"""You are PomeGuard AI Agronomist -- an expert agricultural scientist
37 specialising in fruit health diagnostics, phytochemistry, and
38 precision agronomy. You are currently reviewing a completed fruit
39 analysis report with the farmer or agronomist who submitted it.
40
41 Here is the full context of the analysis you are reviewing:
42
43 FRUIT ANALYSIS REPORT CONTEXT:
44 - Fruit Analysed: {fruit_name}
45 - Date of Analysis: {created_at}
46 - Scan Type: {scan_type}
47
48 DISEASE DETECTION RESULTS:
49 - Detected Condition: {disease_name}
50 - Confidence: {confidence}%
51 - Severity: {severity}
52
53 ENVIRONMENTAL CONDITIONS AT TIME OF ANALYSIS:
54 - UV Irradiance: {uv} W/m2
55 - Humidity: {hum}%
56 - Temperature: {temp}oC
57
58 PHYTOCHEMICAL PROFILE:
59 {phyto_str}
60
61 NUTRITIONAL SCORE: {nut_score} / 100
62 QUALITY GRADE: {grade}
63
64 AGRONOMIC ADVISORY FROM REPORT:
65 {advisory}
66
67 ---
68
69 BEHAVIOUR RULES:
70 - Answer all questions strictly in the context of this specific
71   fruit analysis and the data above
72 - Be conversational but precise -- explain results in plain language
73   a farmer can understand, not academic jargon
74 - If asked about the disease, explain it practically: what it is,
75   how serious it is given the confidence and severity, and what
76   the farmer should do
77 - If asked about phytochemicals, explain what each compound means
78   for the fruit's quality, shelf life, and human health benefit
79 - If asked what to do next, give specific advice derived from the
80   actual values above -- never generic filler
81 - If the user asks something outside this analysis, respond:
82   "I am here specifically to help you understand your PomeGuard
83   analysis results. Do you have any questions about your
84   {fruit_name} scan?"

```

```
85 - Never fabricate values or reference data not present above
86 - Keep responses to 3-5 sentences unless the user asks for more
87 - Do not use markdown in responses -- plain conversational text only
88 ---"""
89
90 async def stream_chat_message(system_prompt: str, history: list, user_message: str):
91     """Async generator to stream response text natively from Gemini Flash 2.5"""
92     model = genai.GenerativeModel(
93         model_name="gemini-3-flash-preview",
94         system_instruction=system_prompt
95     )
96
97     gemini_history = [
98         {"role": msg["role"], "parts": [msg["content"]]}
99         for msg in history
100     ]
101
102     chat = model.start_chat(history=gemini_history)
103     response = chat.send_message(user_message, stream=True)
104
105     for chunk in response:
106         text = chunk.text
107         if text:
108             yield text
109
110 def verify_image_is_pomegranate(image_path: str) -> bool:
111     """Uses Gemini Vision to check if the uploaded image actually contains a pomegranate fruit/plant/leaf."""
112     try:
113         model = genai.GenerativeModel("gemini-3-flash-preview")
114         import PIL.Image
115         img = PIL.Image.open(image_path)
116         prompt = "Look at this image very carefully. Does this image clearly show a pomegranate fruit,
117 pomegranate plant, or pomegranate leaf? Answer strictly with a single word: YES or NO."
118         response = model.generate_content([prompt, img])
119         text = response.text.strip().upper()
120         # If it says YES, or if we can't get a clear NO (to be safe and not block edge cases)
121         if "YES" in text:
122             return True
123         elif "NO" in text:
124             return False
125         return True # Fallback
126     except Exception as e:
127         print(f"Gemini verification failed: {e}")
128         return True # Fallback to true if API fails so we don't break the app
```

2.5 Firebase Administrative Layer (api/utils/firebase_admin.py)

Relative Path: api/utils/firebase_admin.py

```

1 import firebase_admin
2 from firebase_admin import credentials, firestore
3 import os
4 import uuid
5 from datetime import datetime, timedelta
6
7 def get_firestore_client():
8     if not firebase_admin._apps:
9         try:
10             private_key = os.getenv("FIREBASE_ADMIN_PRIVATE_KEY", "").replace("\\n", "\n")
11             project_id = os.getenv("FIREBASE_ADMIN_PROJECT_ID")
12             client_email = os.getenv("FIREBASE_ADMIN_CLIENT_EMAIL")
13
14             if project_id and client_email and private_key:
15                 cred = credentials.Certificate({
16                     "type": "service_account",
17                     "project_id": project_id,
18                     "private_key": private_key,
19                     "client_email": client_email,
20                     "token_uri": "https://oauth2.googleapis.com/token"
21                 })
22                 firebase_admin.initialize_app(cred)
23             else:
24                 return None
25         except Exception as e:
26             print(f"Firebase Init Error (Keys missing?): {e}")
27             return None
28
29     try:
30         return firestore.client()
31     except ValueError:
32         return None
33
34 def get_or_create_session(user_id: str, scan_id: str) -> dict:
35     db = get_firestore_client()
36     if not db:
37         return {} # Fallback graceful failure if no firebase found
38
39     doc_ref = db.collection("chatSessions").document(f"{user_id}_{scan_id}")
40     doc = doc_ref.get()
41
42     if doc.exists:
43         return doc.to_dict()
44
45     now_utc = datetime.utcnow()
46     session_data = {
47         "sessionId": str(uuid.uuid4()),
48         "userId": user_id,
49         "scanId": scan_id,
50         "createdAt": now_utc,
51         "expiresAt": now_utc + timedelta(hours=24)
52     }
53
54     try:
55         doc_ref.set(session_data)
56     except Exception as e:
57         print(f"Firestore Session Creation Error: {e}")
58
59     return session_data
60
61 def save_message(user_id: str, scan_id: str, role: str, content: str):
62     db = get_firestore_client()
63     if not db:
64         return
65
66     doc_ref = db.collection("chatSessions").document(f"{user_id}_{scan_id}").collection("messages").document()
67     try:
68         doc_ref.set({
69             "role": role,
70             "content": content,
71             "timestamp": firestore.SERVER_TIMESTAMP
72         })
73     except Exception as e:
74         print(f"Firestore Save Message Error: {e}")
75
76 def get_session_messages(user_id: str, scan_id: str) -> list:
77     db = get_firestore_client()
78     if not db:
79         return []
80
81     try:
82         messages_ref = db.collection("chatSessions").document(f"{user_id}_{scan_id}").collection("messages")
83         docs = messages_ref.order_by("timestamp").stream()
84

```

```
85     # We explicitly assemble a list mapping {"role": obj, "content": obj}
86     return [{"role": doc.to_dict().get("role"), "content": doc.to_dict().get("content")} for doc in docs]
87 except Exception as e:
88     print(f"Firestore Get Messages Error: {e}")
89     return []
```

3 React Frontend Client

3.1 Vite React Client Entrypoint (frontend/src/main.jsx)

Relative Path: frontend/src/main.jsx

```
1 import React from 'react'
2 import ReactDOM from 'react-dom/client'
3 import { BrowserRouter } from 'react-router-dom'
4 import App from './App.jsx'
5 import './index.css'
6
7 ReactDOM.createRoot(document.getElementById('root')).render(
8   <React.StrictMode>
9     <BrowserRouter>
10       <App />
11     </BrowserRouter>
12   </React.StrictMode>,
13 )
```


3.2 React Routing & State Management (frontend/src/App.jsx)

Relative Path: frontend/src/App.jsx

```

1  import { Routes, Route, Navigate, useLocation } from "react-router-dom"
2  import { useEffect } from "react"
3  import Layout from "../components/Layout"
4  import Home from "../pages/Home"
5  import Analyse from "../pages/Analyse"
6  import Dashboard from "../pages/Dashboard"
7  import HowItWorks from "../pages/HowItWorks"
8  import Login from "../pages/Login"
9  import Signup from "../pages/Signup"
10 import History from "../pages/History"
11 import ScanDetail from "../pages/ScanDetail"
12 import { useAppStore } from "@store/useAppStore"
13 import { Toaster } from "sonner"
14
15 const ProtectedRoute = ({ children }) => {
16   const { session } = useAppStore()
17   const location = useLocation()
18   if (!session) {
19     return <Navigate to={`/login?redirect=${location.pathname}`} replace />
20   }
21   return children
22 }
23
24 const PublicRoute = ({ children }) => {
25   const { session } = useAppStore()
26   if (session) {
27     return <Navigate to="/dashboard" replace />
28   }
29   return children
30 }
31
32 function App() {
33   const { initializeAuth } = useAppStore()
34
35   useEffect(() => {
36     initializeAuth()
37   }, [initializeAuth])
38
39   return (
40     <>
41       <Routes>
42         <Route path="/" element={<Layout />} />
43         <Route index element={<Home />} />
44         <Route path="login" element={<PublicRoute><Login /></PublicRoute>} />
45         <Route path="signup" element={<PublicRoute><Signup /></PublicRoute>} />
46         <Route path="how-it-works" element={<HowItWorks />} />
47         <Route path="analyse" element={<ProtectedRoute><Analyse /></ProtectedRoute>} />
48         <Route path="dashboard" element={<ProtectedRoute><Dashboard /></ProtectedRoute>} />
49         <Route path="history" element={<ProtectedRoute><History /></ProtectedRoute>} />
50         <Route path="scan/:id" element={<ProtectedRoute><ScanDetail /></ProtectedRoute>} />
51       </Routes>
52       <Toaster position="bottom-right" richColors />
53     </>
54   )
55 }
56
57 export default App

```

3.3 Navigation Control Component (frontend/src/components/Navbar.jsx)

Relative Path: frontend/src/components/Navbar.jsx

```

1  import { Link, useLocation, useNavigate } from "react-router-dom"
2  import { useEffect, useState, useRef } from "react"
3  import { useAppStore } from "@store/useAppStore"
4  import { Moon, Sun, Menu, Bell, CheckCircle2, LogOut, User, History, ChevronDown } from "lucide-react"
5  import { Button } from "./ui/button"
6  import { getNotifications, markNotificationAsRead } from "@services/api"
7  import { supabase } from "@lib/supabaseClient"
8  import { toast } from "sonner"
9
10 export default function Navbar() {
11   const { theme, toggleTheme, notifications, setNotifications, markNotificationRead, user, profile } =
     useAppStore()
12   const location = useLocation()
13   const navigate = useNavigate()
14   const [showNotifications, setShowNotifications] = useState(false)
15   const [showUserMenu, setShowUserMenu] = useState(false)
16   const lastUnreadCount = useRef(0)
17   const notifRef = useRef(null)
18   const userMenuRef = useRef(null)
19
20   // Close dropdowns on outside click
21   useEffect(() => {
22     const handler = (e) => {
23       if (notifRef.current && !notifRef.current.contains(e.target)) setShowNotifications(false)
24       if (userMenuRef.current && !userMenuRef.current.contains(e.target)) setShowUserMenu(false)
25     }
26     document.addEventListener("mousedown", handler)
27     return () => document.removeEventListener("mousedown", handler)
28   }, [])
29
30   useEffect(() => {
31     if (!user) return
32     const fetchNotifs = async () => {
33       try {
34         const data = await getNotifications()
35         setNotifications(data)
36         const unreads = data.filter(n => !n.read)
37         if (unreads.length > lastUnreadCount.current) {
38           const newest = unreads[0]
39           if (newest) {
40             toast(newest.type === 'disease_alert' ? "Disease Alert" : "Environmental Alert", {
41               description: newest.message,
42               icon: <Bell className="h-4 w-4" />,
43             })
44           }
45         }
46         lastUnreadCount.current = unreads.length
47       } catch (error) {
48         console.error("Failed to fetch notifications", error)
49       }
50     }
51     fetchNotifs()
52     const interval = setInterval(fetchNotifs, 5000)
53     return () => clearInterval(interval)
54   }, [user, setNotifications])
55
56   const unreadCount = notifications.filter(n => !n.read).length
57
58   const handleMarkAsRead = async (id) => {
59     try {
60       await markNotificationAsRead(id)
61       markNotificationRead(id)
62     } catch (e) {
63       console.error(e)
64     }
65   }
66
67   const handleLogout = async () => {
68     await supabase.auth.signOut()
69     toast.success("Logged out")
70     navigate("/")
71   }
72
73   const publicLinks = [
74     { name: "Home", path: "/" },
75     { name: "How It Works", path: "/how-it-works" },
76   ]
77
78   const authLinks = [
79     { name: "Analyse", path: "/analyse" },
80     { name: "Dashboard", path: "/dashboard" },
81     { name: "History", path: "/history" },
82   ]
83

```

```

84  const links = user ? [...publicLinks, ...authLinks] : publicLinks
85
86  return (
87    <nav className="sticky top-0 z-50 w-full border-b bg-background/95 backdrop-blur supports-[backdrop-filter]:bg-background/60 text-foreground">
88      <div className="container flex h-16 items-center justify-between">
89        <Link to="/" className="flex items-center space-x-2">
90          <div className="flex items-center justify-center bg-primary rounded-full w-8 h-8">
91            <svg xmlns="http://www.w3.org/2000/svg" viewBox="0 0 24 24" fill="none" stroke="currentColor"
strokeWidth="2" strokeLinecap="round" strokeLinejoin="round" className="h-5 w-5 text-white"><circle cx="
12" cy="12" r="10"/><path d="M12 2v20"/><path d="M12 12 4.5 7.5"/><path d="M12 12 17.5 4.5"/></svg>
92          </div>
93          <span className="font-bold text-xl text-primary tracking-tight">PomeGuard</span>
94        </Link>
95
96        <div className="hidden md:flex items-center space-x-6 text-sm font-medium">
97          {links.map((link) => (
98            <Link
99              key={link.path}
100              to={link.path}
101              className={`transition-colors py-5 hover:text-primary ${
102                location.pathname === link.path ? "text-primary border-b-2 border-primary" : "text-muted-
foreground"
103              }`}
104            >
105              {link.name}
106            </Link>
107          ))}
108        </div>
109
110        <div className="flex items-center space-x-2">
111          {/* Notification Bell -- only for logged-in users */}
112          {user && (
113            <div className="relative" ref={notifRef}>
114              <Button variant="ghost" size="icon" onClick={() => setShowNotifications(!showNotifications)}>
115                <Bell className="h-5 w-5" />
116                {unreadCount > 0 && (
117                  <span className="absolute top-1 right-1 h-2.5 w-2.5 rounded-full bg-destructive border
border-background"></span>
118                )}
119              </Button>
120
121              {showNotifications && (
122                <div className="absolute right-0 mt-2 w-80 bg-background border rounded-md shadow-lg overflow-
hidden z-50">
123                  <div className="p-3 border-b bg-muted/50 flex justify-between items-center">
124                    <h3 className="font-semibold text-sm">Notifications</h3>
125                    <span className="text-xs text-muted-foreground">{unreadCount} unread</span>
126                  </div>
127                  <div className="max-h-[300px] overflow-y-auto">
128                    {notifications.length === 0 ? (
129                      <div className="p-4 text-center text-sm text-muted-foreground">No notifications yet</div>
130                    ) : (
131                      notifications.map(notification => (
132                        <div key={notification.id} className={`p-4 border-b text-sm transition-colors ${
notification.read ? "bg-background opacity-70" : "bg-muted/20"}`}>
133                          <div className="flex justify-between items-start gap-2">
134                            <div className="space-y-1">
135                              <p className="font-medium leading-none">{notification.type === 'disease_alert' ?
"Disease Alert" : "Weather Alert"}</p>
136                              <p className="text-xs text-muted-foreground">{notification.message}</p>
137                              <p className="text-[10px] text-muted-foreground">{new Date(notification.
timestamp).toLocaleString()}</p>
138                            </div>
139                            {!notification.read && (
140                              <Button variant="ghost" size="icon" className="h-6 w-6 shrink-0" onClick={() =>
handleMarkAsRead(notification.id)}>
141                                <CheckCircle2 className="h-4 w-4" />
142                              </Button>
143                            )}
144                          </div>
145                        </div>
146                      ))
147                    )}
148                  </div>
149                </div>
150              )}
151            </div>
152          )}
153
154          <Button variant="ghost" size="icon" onClick={toggleTheme}>
155            {theme === "light" ? <Moon className="h-5 w-5" /> : <Sun className="h-5 w-5" />}
156          </Button>
157
158          {/* Auth section */}
159          {user ? (
160            <div className="relative" ref={userMenuRef}>
161              <Button
162                variant="outline"

```

```

163         size="sm"
164         className="flex items-center gap-2 rounded-full pl-2 pr-3"
165         onClick={() => setShowUserMenu(!showUserMenu)}
166     >
167     <div className="h-6 w-6 rounded-full bg-primary/20 flex items-center justify-center">
168     <User className="h-3.5 w-3.5 text-primary" />
169     </div>
170     <span className="text-sm font-medium max-w-[100px] truncate hidden sm:block">
171     {profile?.full_name || user.email?.split('@')[0]}
172     </span>
173     <ChevronDown className="h-3.5 w-3.5 text-muted-foreground" />
174 </Button>
175
176     {showUserMenu && (
177     <div className="absolute right-0 mt-2 w-52 bg-background border rounded-md shadow-lg overflow-
178     hidden z-50">
179         <div className="px-4 py-3 border-b bg-muted/30">
180             <p className="text-sm font-medium leading-none">{profile?.full_name || "User"}</p>
181             <p className="text-xs text-muted-foreground mt-1 truncate">{user.email}</p>
182         </div>
183         <div className="py-1">
184             <Link to="/history" onClick={() => setShowUserMenu(false)} className="flex items-center
185             gap-2 px-4 py-2 text-sm hover:bg-muted transition-colors">
186                 <History className="h-4 w-4" /> Scan History
187             </Link>
188             <Link to="/dashboard" onClick={() => setShowUserMenu(false)} className="flex items-center
189             gap-2 px-4 py-2 text-sm hover:bg-muted transition-colors">
190                 <User className="h-4 w-4" /> Dashboard
191             </Link>
192             <button onClick={handleLogout} className="w-full flex items-center gap-2 px-4 py-2 text-sm
193             text-destructive hover:bg-destructive/10 transition-colors">
194                 <LogOut className="h-4 w-4" /> Sign Out
195             </button>
196         </div>
197     </div>
198     )}
199 </div>
200 ) : (
201     <div className="hidden md:flex items-center gap-2">
202         <Link to="/login">
203             <Button variant="ghost" size="sm">Login</Button>
204         </Link>
205         <Link to="/signup">
206             <Button size="sm" className="rounded-full">Sign Up</Button>
207         </Link>
208     </div>
209 )}
210
211 <Button variant="ghost" size="icon" className="md:hidden">
212     <Menu className="h-5 w-5" />
213 </Button>
214 </div>
215 </nav>
216 )
217 }

```

3.4 Marketing Landing View (frontend/src/pages/Home.jsx)

Relative Path: frontend/src/pages/Home.jsx

```

1  import { motion } from "framer-motion"
2  import { Link } from "react-router-dom"
3  import { Sun, ArrowRight, Microscope, Database } from "lucide-react"
4  import { Button } from "@components/ui/button"
5
6  const stats = [
7    { value: "12,000+", label: "Fruits Analysed" },
8    { value: "99.5%", label: "Detection Accuracy" },
9    { value: "5", label: "Disease Categories" },
10   { value: "3", label: "Districts Covered" },
11 ]
12
13 const features = [
14   {
15     icon: <Microscope className="h-8 w-8 text-primary flex-shrink-0" />,
16     title: "Disease Detection",
17     description: "EfficientNetB0 classifies 5 disease categories with 99.5% accuracy.",
18   },
19   {
20     icon: <Database className="h-8 w-8 text-primary flex-shrink-0" />,
21     title: "Nutritional Inference",
22     description: "OWL/RDF ontology maps disease severity to internal phytochemical state.",
23   },
24   {
25     icon: <Sun className="h-8 w-8 text-primary flex-shrink-0" />,
26     title: "Environmental Intelligence",
27     description: "Real-time UV, humidity, and temperature stress scoring.",
28   },
29 ]
30
31 export default function Home() {
32   return (
33     <div className="flex flex-col min-h-screen">
34       <div className="flex flex-col min-h-screen">
35         <div className="flex flex-col min-h-screen">
36           <div className="flex flex-col min-h-screen">
37             <div className="flex flex-col min-h-screen">
38               <div className="flex flex-col min-h-screen">
39                 <div className="flex flex-col min-h-screen">
40                   <div className="flex flex-col min-h-screen">
41                     <div className="flex flex-col min-h-screen">
42                       <div className="flex flex-col min-h-screen">
43                         <div className="flex flex-col min-h-screen">
44                           <div className="flex flex-col min-h-screen">
45                             <div className="flex flex-col min-h-screen">
46                               <div className="flex flex-col min-h-screen">
47                                 <div className="flex flex-col min-h-screen">
48                                   <div className="flex flex-col min-h-screen">
49                                     <div className="flex flex-col min-h-screen">
50                                       <div className="flex flex-col min-h-screen">
51                                         <div className="flex flex-col min-h-screen">
52                                           <div className="flex flex-col min-h-screen">
53                                             <div className="flex flex-col min-h-screen">
54                                               <div className="flex flex-col min-h-screen">
55                                                 <div className="flex flex-col min-h-screen">
56                                                   <div className="flex flex-col min-h-screen">
57                                                     <div className="flex flex-col min-h-screen">
58                                                       <div className="flex flex-col min-h-screen">
59                                                         <div className="flex flex-col min-h-screen">
60                                                           <div className="flex flex-col min-h-screen">
61                                                             <div className="flex flex-col min-h-screen">
62                                                               <div className="flex flex-col min-h-screen">
63                                                                 <div className="flex flex-col min-h-screen">
64                                                                  <div className="flex flex-col min-h-screen">
65                                                                    <div className="flex flex-col min-h-screen">
66                                                                      <div className="flex flex-col min-h-screen">
67                                                                        <div className="flex flex-col min-h-screen">
68                                                                          <div className="flex flex-col min-h-screen">
69                                                                            <div className="flex flex-col min-h-screen">
70                                                                              <div className="flex flex-col min-h-screen">
71                                                                                <div className="flex flex-col min-h-screen">
72                                                                                  <div className="flex flex-col min-h-screen">
73                                                                                    <div className="flex flex-col min-h-screen">
74                                                                                      <div className="flex flex-col min-h-screen">
75                                                                                        <div className="flex flex-col min-h-screen">
76                                                                                          <div className="flex flex-col min-h-screen">
77                                                                                            <div className="flex flex-col min-h-screen">
78                                                                                              <div className="flex flex-col min-h-screen">
79                                                                                                <div className="flex flex-col min-h-screen">

```

```
80         whileInView={{ opacity: 1, y: 0 }}
81         viewport={{ once: true }}
82         transition={{ duration: 0.6, delay: i * 0.1 }}
83         className="flex flex-col items-center space-y-1"
84       >
85         <h3 className="text-3xl md:text-4xl font-bold tracking-tighter text-foreground">{stat.value}</h3>
86         <p className="text-sm font-medium text-muted-foreground">{stat.label}</p>
87       </motion.div>
88     )}
89   </div>
90 </div>
91 </section>
92
93 { /* Features Section */}
94 <section className="py-24 bg-background flex-1">
95   <div className="container px-4 md:px-6">
96     <div className="grid md:grid-cols-3 gap-8">
97       {features.map((feature, i) => (
98         <motion.div
99           key={feature.title}
100           initial={{ opacity: 0, y: 40 }}
101           whileInView={{ opacity: 1, y: 0 }}
102           viewport={{ once: true }}
103           transition={{ duration: 0.6, delay: i * 0.2 }}
104           className="flex flex-col items-center text-center p-8 rounded-2xl bg-card border shadow-sm
105           hover:shadow-md transition-all"
106         >
107           <div className="mb-6 p-4 rounded-full bg-primary/10">
108             {feature.icon}
109           </div>
110           <h3 className="text-xl font-bold mb-3 text-foreground">{feature.title}</h3>
111           <p className="text-muted-foreground leading-relaxed">
112             {feature.description}
113           </p>
114         </motion.div>
115       ))}
116     </div>
117   </div>
118 </section>
119 </div>
120 )
}
```

3.5 Instructional Flow View (frontend/src/pages/HowItWorks.jsx)

Relative Path: frontend/src/pages/HowItWorks.jsx

```

1  import { motion } from "framer-motion"
2  import { Camera, Cpu, Network, FileText } from "lucide-react"
3
4  const stages = [
5    {
6      icon: <Camera className="h-6 w-6" />,
7      title: "Image Capture",
8      description: "Photograph your Bhagwa pomegranate at 30cm distance under natural light. This non-
9        destructive visual signal serves as the entire foundation of the analysis.",
10    },
11    {
12      icon: <Cpu className="h-6 w-6" />,
13      title: "AI Classification",
14      description: "Our EfficientNetB0 vision model identifies the disease type and severity across 5 bespoke
15        categories trained specifically on Solapur orchard data.",
16    },
17    {
18      icon: <Network className="h-6 w-6" />,
19      title: "Ontology Reasoning",
20      description: "An OWL/RDF knowledge base infers the internal phytochemical concentrations (Anthocyanins,
21        Punicalagins) from the external visual damage marks.",
22    },
23    {
24      icon: <FileText className="h-6 w-6" />,
25      title: "Scorecard Generation",
26      description: "LangGraph agents synthesise a full Nutritional & Health Scorecard along with targeted
27        agronomic advisory for the farmer.",
28    },
29  ],
30
31  export default function HowItWorks() {
32    return (
33      <div className="container max-w-4xl py-12 md:py-24 animate-in fade-in duration-500">
34        <div className="text-center space-y-4 mb-16">
35          <h1 className="text-4xl md:text-5xl font-extrabold tracking-tight">How PomeGuard Works</h1>
36          <p className="text-xl text-muted-foreground max-w-2xl mx-auto">
37            Understanding the invisible logic behind Maharashtra's leading non-destructive quality assessment.
38          </p>
39        </div>
40
41        <div className="relative space-y-12 before:absolute before:inset-0 before:ml-8 before:-translate-x-px md:
42          :before:mx-auto md:before:translate-x-0 before:h-full before:w-1 before:bg-gradient-to-b before:from-
43          transparent before:via-border before:to-transparent">
44          {stages.map((stage, index) => (
45            <motion.div
46              key={index}
47              initial={{ opacity: 0, y: 50 }}
48              whileInView={{ opacity: 1, y: 0 }}
49              viewport={{ once: true, margin: "-100px" }}
50              transition={{ duration: 0.6 }}
51              className="relative flex items-center justify-between md:justify-normal md:odd:flex-row-reverse
52            group">
53              <div className="flex items-center justify-center w-16 h-16 rounded-full border-4 border-background
54                bg-primary/10 text-primary shadow-sm z-10 shrink-0 md:order-1 md:hidden">
55                {stage.icon}
56              </div>
57
58              <div className="hidden md:flex items-center justify-center w-16 h-16 rounded-full border-4 border-
59                background bg-primary/10 text-primary shadow-sm z-10 shrink-0 md:absolute md:left-1/2 md:-translate-x
60                -1/2">
61                {stage.icon}
62              </div>
63
64              <div className="w-[calc(100%-5rem)] md:w-[calc(50%-3rem)] p-6 rounded-2xl bg-card border shadow-sm
65                transition-shadow hover:shadow-md">
66                <h3 className="text-xl font-bold mb-2 text-foreground">{stage.title}</h3>
67                <p className="text-muted-foreground leading-relaxed">{stage.description}</p>
68              </div>
69            </motion.div>
70          )))
71        </div>
72
73        <div className="mt-24 pt-12 border-t text-center space-y-8">
74          <h2 className="text-2xl font-bold">Powered by Modern Technology</h2>
75          <div className="flex flex-wrap justify-center gap-3">
76            {[ "Python", "EfficientNetB0", "OWL/RDF", "LangGraph", "FastAPI", "React", "Zustand" ].map((tech) => (
77              <span key={tech} className="px-4 py-2 rounded-full border bg-muted/30 text-sm font-medium text-
78                foreground">
79                {tech}
80              </span>
81            ))}
82          </div>
83          <p className="text-muted-foreground mx-auto pt-4 leading-relaxed max-w-3xl">

```

```
73         Bhagwa pomegranate cultivation in Maharashtra drives significant agricultural export value. Non-  
       destructive quality assessment prevents millions in lost revenue by ensuring accurate grading without  
       sacrificing the crop.  
74     </p>  
75 </div>  
76 </div>  
77 )  
78 }
```


3.6 Authentication: Login Screen (frontend/src/pages/Login.jsx)

Relative Path: frontend/src/pages/Login.jsx

```

1  import { useState } from 'react'
2  import { Link, useNavigate } from 'react-router-dom'
3  import { supabase } from '@lib/supabaseClient'
4  import { Button } from '@components/ui/button'
5  import { toast } from 'sonner'
6
7  export default function Login() {
8    const [email, setEmail] = useState('')
9    const [password, setPassword] = useState('')
10   const [loading, setLoading] = useState(false)
11   const navigate = useNavigate()
12
13   const handleLogin = async (e) => {
14     e.preventDefault()
15     setLoading(true)
16     const { error } = await supabase.auth.signInWithPassword({ email, password })
17     setLoading(false)
18
19     if (error) {
20       toast.error(error.message)
21     } else {
22       toast.success('Logged in successfully!')
23       navigate('/dashboard')
24     }
25   }
26
27   return (
28     <div className="flex items-center justify-center min-h-[calc(100vh-4rem)]">
29       <div className="w-full max-w-md p-8 space-y-6 bg-card rounded-lg border shadow-sm">
30         <div className="text-center">
31           <h1 className="text-2xl font-bold tracking-tight">Welcome back</h1>
32           <p className="text-sm text-muted-foreground mt-2">Login to access your dashboard and history.</p>
33         </div>
34         <form onSubmit={handleLogin} className="space-y-4">
35           <div className="space-y-2">
36             <label className="text-sm font-medium">Email</label>
37             <input
38               type="email"
39               required
40               value={email}
41               onChange={(e) => setEmail(e.target.value)}
42               className="w-full p-2 border rounded-md bg-background"
43             />
44           </div>
45           <div className="space-y-2">
46             <label className="text-sm font-medium">Password</label>
47             <input
48               type="password"
49               required
50               value={password}
51               onChange={(e) => setPassword(e.target.value)}
52               className="w-full p-2 border rounded-md bg-background"
53             />
54           </div>
55           <Button type="submit" className="w-full" disabled={loading}>
56             {loading ? 'Logging in...' : 'Login'}
57           </Button>
58         </form>
59         <div className="text-center text-sm">
60           Don't have an account? <Link to="/signup" className="text-primary hover:underline">Sign up</Link>
61         </div>
62       </div>
63     </div>
64   )
65 }

```

3.7 Authentication: Registration Screen (frontend/src/pages/Signup.jsx)

Relative Path: frontend/src/pages/Signup.jsx

```

1  import { useState } from 'react'
2  import { Link, useNavigate } from 'react-router-dom'
3  import { supabase } from '@lib/supabaseClient'
4  import { Button } from '@components/ui/button'
5  import { toast } from 'sonner'
6
7  export default function Signup() {
8    const [fullName, setFullName] = useState('')
9    const [email, setEmail] = useState('')
10   const [password, setPassword] = useState('')
11   const [confirmPassword, setConfirmPassword] = useState('')
12   const [loading, setLoading] = useState(false)
13   const navigate = useNavigate()
14
15   const handleSignup = async (e) => {
16     e.preventDefault()
17     if (password !== confirmPassword) {
18       toast.error('Passwords do not match')
19       return
20     }
21
22     setLoading(true)
23     const { error } = await supabase.auth.signUp({
24       email,
25       password,
26       options: { data: { full_name: fullName } }
27     })
28     setLoading(false)
29
30     if (error) {
31       toast.error(error.message)
32     } else {
33       toast.success('Account created! Please login.')
34       navigate('/dashboard')
35     }
36   }
37
38   return (
39     <div className="flex items-center justify-center min-h-[calc(100vh-4rem)]">
40       <div className="w-full max-w-md p-8 space-y-6 bg-card rounded-lg border shadow-sm">
41         <div className="text-center">
42           <h1 className="text-2xl font-bold tracking-tight">Create an account</h1>
43           <p className="text-sm text-muted-foreground mt-2">Sign up to manage your scans and crops.</p>
44         </div>
45         <form onSubmit={handleSignup} className="space-y-4">
46           <div className="space-y-2">
47             <label className="text-sm font-medium">Full Name</label>
48             <input
49               type="text"
50               required
51               value={fullName}
52               onChange={e => setFullName(e.target.value)}
53               className="w-full p-2 border rounded-md bg-background"
54             />
55           </div>
56           <div className="space-y-2">
57             <label className="text-sm font-medium">Email</label>
58             <input
59               type="email"
60               required
61               value={email}
62               onChange={e => setEmail(e.target.value)}
63               className="w-full p-2 border rounded-md bg-background"
64             />
65           </div>
66           <div className="space-y-2">
67             <label className="text-sm font-medium">Password</label>
68             <input
69               type="password"
70               required
71               value={password}
72               onChange={e => setPassword(e.target.value)}
73               className="w-full p-2 border rounded-md bg-background"
74             />
75           </div>
76           <div className="space-y-2">
77             <label className="text-sm font-medium">Confirm Password</label>
78             <input
79               type="password"
80               required
81               value={confirmPassword}
82               onChange={e => setConfirmPassword(e.target.value)}
83               className="w-full p-2 border rounded-md bg-background"
84             />

```

```
85         </div>
86         <Button type="submit" className="w-full" disabled={loading}>
87           {loading ? 'Signing up...' : 'Sign up'}
88         </Button>
89       </form>
90       <div className="text-center text-sm">
91         Already have an account? <Link to="/login" className="text-primary hover:underline">Login</Link>
92       </div>
93     </div>
94   </div>
95 )
96 }
```

3.8 AI Image Classification & Disease Scanning View (frontend/src/pages/Analyse.jsx)

Relative Path: frontend/src/pages/Analyse.jsx

```

1  import { useState, useRef, useEffect } from "react"
2  import { useNavigate } from "react-router-dom"
3  import { motion, AnimatePresence } from "framer-motion"
4  import { UploadCloud, CheckCircle2, Download, RefreshCw, Leaf, MapPin, Camera, X } from "lucide-react"
5  import { Button } from "@/components/ui/button"
6  import { Card, CardContent, CardHeader, CardTitle } from "@/components/ui/card"
7  import { Badge } from "@/components/ui/badge"
8  import { classifyFruit, submitEnvMetadata, runOntologyInference, uploadMedia, saveScan } from "@/services/api"
9  import { toast } from "sonner"
10
11  export default function Analyse() {
12    const [imageFile, setImageFile] = useState(null)
13    const [preview, setPreview] = useState(null)
14    const [mediaInfo, setMediaInfo] = useState(null)
15    const navigate = useNavigate()
16
17    const [isCameraActive, setIsCameraActive] = useState(false)
18    const [cameraError, setCameraError] = useState(null)
19    const videoRef = useRef(null)
20    const canvasRef = useRef(null)
21
22    const [isClassifying, setIsClassifying] = useState(false)
23    const [diseaseData, setDiseaseData] = useState(null)
24
25    const [envParams, setEnvParams] = useState({ uv_irradiance: 800, humidity: 60, temperature: 25 })
26    const [isSubmittingEnv, setIsSubmittingEnv] = useState(false)
27    const [envData, setEnvData] = useState(null)
28    const [location, setLocation] = useState(null)
29
30    const [isInferring, setIsInferring] = useState(false)
31    const [nutritionData, setNutritionData] = useState(null)
32
33    const [isFetchingWeather, setIsFetchingWeather] = useState(false)
34    const [weatherError, setWeatherError] = useState(null)
35
36    const [isFetchingLocation, setIsFetchingLocation] = useState(false)
37    const [locationError, setLocationError] = useState(null)
38
39    const detectLocation = () => {
40      setIsFetchingLocation(true)
41      setLocationError(null)
42
43      if (!navigator.geolocation) {
44        setLocationError("Geolocation is not supported by your browser")
45        setIsFetchingLocation(false)
46        return
47      }
48
49      navigator.geolocation.getCurrentPosition(
50        (position) => {
51          const { latitude, longitude } = position.coords
52          setLocation({ latitude, longitude })
53          setIsFetchingLocation(false)
54        },
55        (error) => {
56          setLocationError("Location access denied or unavailable.")
57          setIsFetchingLocation(false)
58        }
59      )
60    }
61
62    const autoFetchWeather = () => {
63      setIsFetchingWeather(true)
64      setWeatherError(null)
65
66      if (!navigator.geolocation) {
67        setWeatherError("Geolocation is not supported by your browser")
68        setIsFetchingWeather(false)
69        return
70      }
71
72      navigator.geolocation.getCurrentPosition(
73        async (position) => {
74          try {
75            const { latitude, longitude } = position.coords
76            setLocation({ latitude, longitude })
77            const res = await fetch(`https://api.open-meteo.com/v1/forecast?latitude=${latitude}&longitude=${longitude}&current=temperature_2m,relative_humidity_2m,shortwave_radiation&timezone=auto`)
78            const data = await res.json()
79
80            if (data?.current) {
81              setEnvParams(prev => ({
82                ...prev,
83                temperature: data.current.temperature_2m,

```

```

84         humidity: data.current.relative_humidity_2m,
85         uv_irradiance: Math.round(data.current.shortwave_radiation)
86     )))
87     } else {
88         setWeatherError("Failed to parse weather data")
89     }
90     } catch (err) {
91         setWeatherError("Failed to fetch weather data")
92         console.error(err)
93     } finally {
94         setIsFetchingWeather(false)
95     }
96 },
97 (error) => {
98     setWeatherError("Location access denied or unavailable.")
99     setIsFetchingWeather(false)
100 }
101 )
102 }
103
104 const startCamera = async () => {
105     setIsCameraActive(true)
106     setCameraError(null)
107     try {
108         const stream = await navigator.mediaDevices.getUserMedia({
109             video: { facingMode: 'environment' }
110         })
111         if (videoRef.current) {
112             videoRef.current.srcObject = stream
113         }
114     } catch (err) {
115         setCameraError("Camera access denied or unavailable.")
116         console.error(err)
117     }
118 }
119
120 const stopCamera = () => {
121     if (videoRef.current && videoRef.current.srcObject) {
122         videoRef.current.srcObject.getTracks().forEach(track => track.stop())
123         videoRef.current.srcObject = null
124     }
125     setIsCameraActive(false)
126 }
127
128 const capturePhoto = async () => {
129     if (!videoRef.current || !canvasRef.current) return
130     const video = videoRef.current
131     const canvas = canvasRef.current
132     canvas.width = video.videoWidth
133     canvas.height = video.videoHeight
134     const ctx = canvas.getContext('2d')
135     ctx.drawImage(video, 0, 0, canvas.width, canvas.height)
136
137     canvas.toBlob(async (blob) => {
138         if (!blob) return
139         const file = new File([blob], `capture-${Date.now()}.jpg`, { type: 'image/jpeg' })
140
141         stopCamera()
142         setImageFile(file)
143         setPreview(URL.createObjectURL(file))
144
145         setIsClassifying(true)
146         try {
147             const uploadRes = await uploadMedia(file)
148             setMediaInfo(uploadRes)
149
150             const result = await classifyFruit(file)
151             setDiseaseData(result)
152         } catch (err) {
153             console.error(err)
154             toast.error("Failed to process or upload image.")
155         } finally {
156             setIsClassifying(false)
157         }
158     }, 'image/jpeg', 0.9)
159 }
160
161 useEffect(() => {
162     return () => {
163         if (videoRef.current && videoRef.current.srcObject) {
164             videoRef.current.srcObject.getTracks().forEach(track => track.stop())
165         }
166     }
167 }, [])
168
169 const handleImageUpload = async (e) => {
170     const file = e.target.files?.[0]
171     if (!file) return
172     setImageFile(file)
173     setPreview(URL.createObjectURL(file))

```

```

174
175     setIsClassifying(true)
176     try {
177         const uploadRes = await uploadMedia(file)
178         setMediaInfo(uploadRes)
179
180         const result = await classifyFruit(file)
181         setDiseaseData(result)
182     } catch (err) {
183         console.error(err)
184         toast.error("Failed to process or upload image.")
185     } finally {
186         setIsClassifying(false)
187     }
188 }
189
190 const handleEnvSubmit = async (e) => {
191     e.preventDefault()
192     setIsSubmittingEnv(true)
193     try {
194         const result = await submitEnvMetadata({
195             uv_irradiance: Number(envParams.uv_irradiance),
196             humidity: Number(envParams.humidity),
197             temperature: Number(envParams.temperature),
198         })
199         setEnvData(result)
200     } catch (err) {
201         console.error(err)
202     } finally {
203         setIsSubmittingEnv(false)
204     }
205 }
206
207 const handleFullAnalysis = async () => {
208     if (!diseaseData || !envData || !mediaInfo) return
209     setIsInferring(true)
210     try {
211         const nutritionRes = await runOntologyInference({
212             disease: diseaseData.disease,
213             severity: diseaseData.severity,
214             stress_factors: envData.stress_factors,
215         })
216         setNutritionData(nutritionRes)
217
218         const fullResult = {
219             disease: diseaseData.disease,
220             severity: diseaseData.severity,
221             nutrition: nutritionRes,
222             environment: envData
223         }
224
225         const savedScan = await saveScan({
226             scan_type: "disease_detection",
227             media_url: mediaInfo.url,
228             public_id: mediaInfo.publicId,
229             media_type: mediaInfo.mediaType,
230             input_data: {
231                 ...envParams,
232                 ...(location ? { latitude: location.latitude, longitude: location.longitude } : {})
233             },
234             result: fullResult,
235             confidence_score: diseaseData.confidence
236         })
237
238         toast.success("Analysis complete and saved.")
239         setTimeout(() => navigate('/history'), 2000)
240
241     } catch (err) {
242         console.error(err)
243         toast.error("Failed to run pipeline or save.")
244     } finally {
245         setIsInferring(false)
246     }
247 }
248
249 const resetAll = () => {
250     setImageFile(null)
251     setPreview(null)
252     setDiseaseData(null)
253     setEnvData(null)
254     setNutritionData(null)
255 }
256
257 const canRunAnalysis = diseaseData && envData
258 const isFinished = nutritionData !== null
259
260 return (
261     <div className="container py-8 md:py-12 animate-in fade-in duration-500">
262         <div className="flex flex-col md:flex-row gap-8 lg:gap-12">
263

```

```

264      { /* Left Panel */
265      <div className="w-full md:w-5/12 space-y-8 flex flex-col no-print">
266        <div>
267          <h1 className="text-3xl font-extrabold tracking-tight mb-2">Fruit Analysis</h1>
268          <p className="text-muted-foreground leading-relaxed">Upload an image and provide environmental
factors to generate a holistic nutritional scorecard.</p>
269        </div>
270
271        <Card className="shadow-sm border-border">
272          <CardHeader className="pb-3"><CardTitle>1. Visual Upload</CardTitle></CardHeader>
273          <CardContent>
274            {isCameraActive ? (
275              <div className="relative overflow-hidden flex flex-col items-center justify-center p-4 border
-2 border-primary/50 bg-primary/5 rounded-xl">
276                {cameraError ? (
277                  <div className="text-center p-4">
278                    <p className="text-destructive mb-4">{cameraError}</p>
279                    <Button variant="outline" onClick={stopCamera}>Cancel</Button>
280                  </div>
281                ) : (
282                  <div className="flex flex-col items-center w-full space-y-4">
283                    <div className="relative w-full max-w-sm aspect-[3/4] sm:aspect-video bg-black rounded-
lg overflow-hidden flex items-center justify-center">
284                      <video ref={videoRef} autoPlay playsInline muted className="w-full h-full object-cover
"></video>
285                      <canvas ref={canvasRef} className="hidden"></canvas>
286                    </div>
287                    <div className="flex gap-4">
288                      <Button variant="outline" onClick={stopCamera}>
289                        <X className="h-4 w-4 mr-2" /> Cancel
290                      </Button>
291                      <Button onClick={capturePhoto}>
292                        <Camera className="h-4 w-4 mr-2" /> Capture Photo
293                      </Button>
294                    </div>
295                  </div>
296                )}
297              </div>
298            ) : (
299              <label className={`relative overflow-hidden flex flex-col items-center justify-center p-8
border-2 border-dashed rounded-xl cursor-pointer transition-all duration-200 ${preview ? 'border-primary
/50 bg-primary/5' : 'border-muted-foreground/30 hover:border-primary/50 hover:bg-muted/50'} `}>
300                <input type="file" className="hidden" accept="image/jpeg, image/png, image/webp" onChange={
handleImageUpload} />
301                {preview ? (
302                  <div className="flex flex-col items-center space-y-4">
303                    <img src={preview} alt="Preview" className="h-32 w-32 object-cover rounded-lg shadow-sm"
/>
304                    <div className="text-sm font-medium text-center">
305                      <p className="text-foreground max-w-[200px] truncate">{imageFile?.name}</p>
306                      <p className="text-muted-foreground">{(imageFile?.size / 1024 / 1024).toFixed(2)} MB</
p>
307                    </div>
308                    <Button type="button" variant="outline" size="sm" onClick={(e) => { e.preventDefault();
startCamera(); }} className="mt-2 text-xs z-10 relative">
309                      <Camera className="h-3 w-3 mr-1.5" /> Retake Photo
310                    </Button>
311                  </div>
312                ) : (
313                  <div className="flex flex-col items-center text-center space-y-4">
314                    <div className="p-4 bg-background rounded-full text-primary border shadow-sm"><
UploadCloud className="h-8 w-8" /></div>
315                    <div>
316                      <p className="font-semibold text-foreground">Drag your fruit image here</p>
317                      <p className="text-sm text-muted-foreground pt-1 mb-4">or click to browse (JPG, PNG,
WEBP)</p>
318                      <Button type="button" variant="secondary" onClick={(e) => { e.preventDefault();
startCamera(); }} className="mt-2 z-10 relative">
319                        <Camera className="h-4 w-4 mr-2" /> Use Camera
320                      </Button>
321                    </div>
322                  </div>
323                )}
324                {isClassifying && (
325                  <div className="absolute inset-0 bg-background/80 backdrop-blur-sm flex flex-col items-
center justify-center rounded-xl z-10 transition-all">
326                    <RefreshCw className="h-8 w-8 animate-spin text-primary mb-3" />
327                    <p className="text-sm font-medium animate-pulse text-foreground">Classifying disease
...</p>
328                  </div>
329                )}
330              </label>
331            )}
332          </CardContent>
333        </Card>
334
335        <Card className="shadow-sm border-border">
336          <CardHeader className="pb-3"><CardTitle>2. Environmental Metadata</CardTitle></CardHeader>
337          <CardContent>
338            <form onSubmit={handleEnvSubmit} className="space-y-5">

```

```

339         <div className="flex flex-col gap-2 bg-muted/40 p-3 rounded-lg border border-border/50">
340             <span className="text-xs font-medium text-muted-foreground hidden sm:block">Use sensors to
fetch your environment data, or just tag your location.</span>
341             <div className="flex gap-2 justify-end">
342                 <Button type="button" variant="outline" size="sm" onClick={detectLocation} disabled={
isFetchingLocation || location} className="h-7 bg-primary/10 hover:bg-primary/20 border-primary/20 text-
primary transition-all rounded-full px-3 overflow-hidden">
343                     {isFetchingLocation ? <RefreshCw className="h-3 w-3 animate-spin mr-1.5" /> : <MapPin
className="h-3 w-3 mr-1.5" />}
344                     <span className="text-[10px] font-bold uppercase tracking-wider">{location ? "Tagged" :
"Tag Location"}</span>
345                 </Button>
346                 <Button type="button" variant="outline" size="sm" onClick={autoFetchWeather} disabled={
isFetchingWeather} className="h-7 bg-primary/10 hover:bg-primary/20 border-primary/20 text-primary
transition-all rounded-full px-3">
347                     {isFetchingWeather ? <RefreshCw className="h-3 w-3 animate-spin mr-1.5" /> : <RefreshCw
className="h-3 w-3 mr-1.5" />}
348                     <span className="text-[10px] font-bold uppercase tracking-wider">{isFetchingWeather ? "
Fetching..." : "Fetch Weather"}</span>
349                 </Button>
350             </div>
351         </div>
352         {location && (
353             <div className="flex items-center gap-1.5 text-xs font-medium text-emerald-600 dark:text-
emerald-400 bg-emerald-50 dark:bg-emerald-950/30 px-2 py-1.5 rounded border border-emerald-100 dark:
border-emerald-800/50 mt-2 mb-1">
354                 <MapPin className="h-3 w-3" />
355                 Location captured for Hotspot Map
356             </div>
357         )}
358         {weatherError && <p className="text-destructive text-xs font-medium px-1 mt-1 -mb-2">{
weatherError}</p>}
359         {locationError && <p className="text-destructive text-xs font-medium px-1 mt-1 -mb-2">{
locationError}</p>}
360         <div className="grid grid-cols-3 gap-4">
361             <div className="space-y-2">
362                 <label className="text-xs font-semibold text-muted-foreground uppercase tracking-wider">
Solar Rad (W/m2)</label>
363                 <input type="number" min="0" max="1400" required value={envParams.uv_irradiance} onChange=
={e => setEnvParams({...envParams, uv_irradiance: e.target.value})} className="flex h-11 w-full rounded-
md border border-input bg-background px-3 py-2 text-sm ring-offset-background transition-colors focus-
visible:outline-none focus-visible:ring-2 focus-visible:ring-primary focus-visible:ring-offset-2" />
364             </div>
365             <div className="space-y-2">
366                 <label className="text-xs font-semibold text-muted-foreground uppercase tracking-wider">
Humidity (%)</label>
367                 <input type="number" min="0" max="100" required value={envParams.humidity} onChange={e =>
setEnvParams({...envParams, humidity: e.target.value})} className="flex h-11 w-full rounded-md border
border-input bg-background px-3 py-2 text-sm ring-offset-background transition-colors focus-visible:
outline-none focus-visible:ring-2 focus-visible:ring-primary focus-visible:ring-offset-2" />
368             </div>
369             <div className="space-y-2">
370                 <label className="text-xs font-semibold text-muted-foreground uppercase tracking-wider">
Temp (oC)</label>
371                 <input type="number" min="0" max="60" required value={envParams.temperature} onChange={e
=> setEnvParams({...envParams, temperature: e.target.value})} className="flex h-11 w-full rounded-md
border border-input bg-background px-3 py-2 text-sm ring-offset-background transition-colors focus-
visible:outline-none focus-visible:ring-2 focus-visible:ring-primary focus-visible:ring-offset-2" />
372             </div>
373         </div>
374         <Button type="submit" disabled={isSubmittingEnv} variant={envData ? "secondary" : "default"}
className="w-full h-11 relative overflow-hidden transition-all">
375             {isSubmittingEnv ? <RefreshCw className="h-4 w-4 animate-spin mr-2" /> : (envData ? "Update
Metadata" : "Submit Environmental Data")}
376         </Button>
377     </form>
378 </CardContent>
379 </Card>
380
381     <Card className={`shadow-sm transition-all duration-300 ${canRunAnalysis ? 'border-primary shadow-
primary/20 bg-primary/5' : 'border-border bg-card opacity-70'} `>
382         <CardContent className="p-6">
383             <Button
384                 onClick={handleFullAnalysis}
385                 disabled={!canRunAnalysis || isInferring || isFinished}
386                 className="w-full h-14 text-lg font-bold shadow-lg"
387             >
388                 {isInferring ? <RefreshCw className="h-5 w-5 animate-spin mr-2" /> : "Run Full Analysis"}
389             </Button>
390         </CardContent>
391     </Card>
392 </div>
393
394 {/* Right Panel */}
395 <div className="w-full md:w-7/12 flex flex-col justify-start">
396     <div className="bg-card border rounded-2xl p-6 md:p-8 shadow-sm flex-1 print:border-none print:
shadow-none print:p-0 relative overflow-hidden min-h-[600px]">
397
398         {!diseaseData && !envData ? (
399             <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} className="absolute inset-0 flex

```



```

flex-col items-center justify-center text-center space-y-4 px-8 text-muted-foreground opacity-60">
400     <Leaf className="h-24 w-24 mb-4 text-border" />
401     <p className="text-xl font-medium text-foreground">Awaiting Parameters</p>
402     <p className="text-sm max-w-sm">The analysis results will systematically populate here as the
PomeGuard pipeline executes.</p>
403     </motion.div>
404     ) : null}
405
406     <div className="space-y-6 relative z-10 print:space-y-8 h-full flex flex-col">
407       <AnimatePresence>
408         {/* Card 1: Disease */}
409         {diseaseData && (
410           <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} className="p-5
border rounded-xl bg-background shadow-sm hover:shadow-md transition-shadow">
411             <h3 className="text-xs font-bold text-muted-foreground uppercase tracking-widest mb-4
border-b pb-2">1. Visual Disease Detection</h3>
412             <div className="flex items-center gap-6">
413               <div className="w-20 h-20 shrink-0 rounded-lg overflow-hidden border bg-muted">
414                 <img src={diseaseData.image_url} alt="Fruit crop" className="w-full h-full object-
cover" />
415               </div>
416               <div className="flex-1 space-y-3">
417                 <div className="flex justify-between items-start">
418                   <div>
419                     <p className="font-bold text-xl text-foreground">{diseaseData.disease.replace(/_/g
, " ")}</p>
420                     <p className="text-sm text-muted-foreground font-medium">Confidence: {diseaseData.
confidence}%</p>
421                   </div>
422                   <Badge variant={diseaseData.severity === "None" ? "success" : diseaseData.severity
=== "Severe" ? "destructive" : "warning"} className="text-sm py-1">{diseaseData.severity} Severity</Badge>
423                 </div>
424                 <div className="w-full bg-muted rounded-full h-2.5 overflow-hidden">
425                   <motion.div initial={{ width: 0 }} animate={{ width: `${diseaseData.confidence}%` }}
transition={{ duration: 1, ease: "easeOut" }} className="bg-primary h-full rounded-full"></motion.div>
426                 </div>
427               </div>
428             </div>
429           </motion.div>
430         )}
431
432         {/* Card 2: Env */}
433         {envData && (
434           <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} transition={{
delay: 0.1 }} className="p-5 border rounded-xl bg-background shadow-sm hover:shadow-md transition-shadow">
435             <h3 className="text-xs font-bold text-muted-foreground uppercase tracking-widest mb-4
border-b pb-2">2. Environmental Stress Profiles</h3>
436             <div className="grid grid-cols-3 gap-4 mb-4">
437               <div className="flex flex-col border rounded-lg p-3 text-center bg-card">
438                 <span className="text-xs text-muted-foreground font-medium mb-1">Solar Radiation</span>
439                 <span>
<span className="font-bold text-xl text-foreground mb-2">{envParams.uv_irradiance}</
span>
440                 <Badge variant={envData.uv_stress_level === "High" ? "destructive" : "success"}
className="mx-auto uppercase tracking-wide text-[10px] sm:w-full justify-center">{envData.uv_stress_level
}</Badge>
441               </div>
442               <div className="flex flex-col border rounded-lg p-3 text-center bg-card">
443                 <span className="text-xs text-muted-foreground font-medium mb-1">Humidity</span>
444                 <span className="font-bold text-xl text-foreground mb-2">{envParams.humidity}%</span>
445                 <Badge variant={envData.humidity_stress_level === "High" ? "destructive" : "success"}
className="mx-auto uppercase tracking-wide text-[10px] sm:w-full justify-center">{envData.
humidity_stress_level}</Badge>
446               </div>
447               <div className="flex flex-col border rounded-lg p-3 text-center bg-card">
448                 <span className="text-xs text-muted-foreground font-medium mb-1">Temperature</span>
449                 <span>
<span className="font-bold text-xl text-foreground mb-2">{envParams.temperature}oC</
span>
450                 <Badge variant={envData.temp_stress_level === "High" ? "destructive" : "success"}
className="mx-auto uppercase tracking-wide text-[10px] sm:w-full justify-center">{envData.
temp_stress_level}</Badge>
451               </div>
452             </div>
453             <p className="text-sm font-medium text-foreground bg-muted/30 p-4 rounded-lg border-l-4
border-primary/50 flex">
454               <span className="flex-1">{envData.stress_factors.join(", ")} -- moderating structural
and phytochemical stability.</span>
455             </p>
456           </motion.div>
457         )}
458
459         {/* Card 3: Scorecard */}
460         {nutritionData && (
461           <motion.div initial={{ opacity: 0, y: 20 }} animate={{ opacity: 1, y: 0 }} transition={{
delay: 0.2 }} className="p-6 border rounded-xl bg-background shadow-[0_4px_20px_-4px_rgba(0,0,0,0.1)]
flex flex-col sm:flex-row gap-8 items-center border-primary/20 relative overflow-hidden">
462             <div className="absolute top-0 right-0 w-32 h-32 bg-primary/5 rounded-full -translate-y
-1/2 translate-x-1/2 blur-2xl pointer-events-none"></div>

```

```

463         <div className="flex flex-col items-center shrink-0">
464             <div className="relative w-36 h-36 flex items-center justify-center">
465                 <svg className="w-full h-full transform -rotate-90 filter drop-shadow-md">
466                     <circle cx="72" cy="72" r="64" fill="transparent" stroke="currentColor" strokeWidth=
"12" className="text-muted-50" />
467                     <motion.circle
468                         cx="72" cy="72" r="64" fill="transparent" stroke="currentColor" strokeWidth="12"
469                         strokeLinecap="round"
470                         strokeDasharray={402.12}
471                         initial={{ strokeDashoffset: 402.12 }}
472                         animate={{ strokeDashoffset: 402.12 - (402.12 * nutritionData.nutritional_score /
100) }}
473                         transition={{ duration: 1.5, ease: "easeOut", delay: 0.4 }}
474                         className={nutritionData.nutritional_score >= 80 ? "text-brand-success" :
nutritionData.nutritional_score >= 60 ? "text-amber-500" : "text-destructive"}
475                     />
476                 </svg>
477                 <div className="absolute inset-0 flex flex-col items-center justify-center">
478                     <span className="text-5xl font-extrabold tracking-tighter text-foreground">{
nutritionData.nutritional_score}</span>
479                     <span className="text-xs text-muted-foreground font-bold tracking-widest uppercase
mt-1">Score</span>
480                 </div>
481             </div>
482             <Badge className="mt-5 px-4 py-1.5 text-sm uppercase tracking-wider" variant={
nutritionData.nutritional_score >= 80 ? "success" : nutritionData.nutritional_score >= 60 ? "warning" : "
destructive"}>
483                 {nutritionData.quality_tier} Tier
484             </Badge>
485         </div>
486
487         <div className="flex-1 w-full space-y-6">
488             <h3 className="text-xs font-bold text-muted-foreground uppercase tracking-widest border-
b pb-2">3. Nutritional & Health Scorecard</h3>
489             <div>
490                 <div className="flex justify-between text-sm mb-1.5"><span className="font-semibold
text-foreground">Anthocyanins</span><span className="font-bold text-foreground">{nutritionData.
anthocyanins} mg/100g</span></div>
491                 <div className="w-full bg-muted rounded-full h-3 overflow-hidden shadow-inner"><motion
.div initial={{ width: 0 }} animate={{ width: `${Math.min(100, (nutritionData.anthocyanins / 20)*100)}%`
}} transition={{ duration: 1, delay: 0.5 }} className="bg-gradient-to-r from-[#5B2C6F] to-[#8E44AD] h-
full rounded-full"></motion.div></div>
492                 </div>
493                 <div>
494                     <div className="flex justify-between text-sm mb-1.5"><span className="font-semibold
text-foreground">Punicalagins</span><span className="font-bold text-foreground">{nutritionData.
punicalagins} mg/100g</span></div>
495                     <div className="w-full bg-muted rounded-full h-3 overflow-hidden shadow-inner"><motion
.div initial={{ width: 0 }} animate={{ width: `${Math.min(100, (nutritionData.punicalagins / 200)*100)}%`
}} transition={{ duration: 1, delay: 0.6 }} className="bg-gradient-to-r from-[#8B1A1A] to-[#C0392B] h-
full rounded-full"></motion.div></div>
496                     </div>
497                     <div>
498                         <div className="flex justify-between text-sm mb-1.5"><span className="font-semibold
text-foreground">Ellagic Acid</span><span className="font-bold text-foreground">{nutritionData.
ellagic_acid} mg/100g</span></div>
499                         <div className="w-full bg-muted rounded-full h-3 overflow-hidden shadow-inner"><motion
.div initial={{ width: 0 }} animate={{ width: `${Math.min(100, (nutritionData.ellagic_acid / 15)*100)}%`
}} transition={{ duration: 1, delay: 0.7 }} className="bg-gradient-to-r from-[#D35400] to-[#E67E22] h-
full rounded-full"></motion.div></div>
500                         </div>
501                     </div>
502                 </motion.div>
503             </div>
504
505             {/* Card 4: Advisory */}
506             {nutritionData && (
507                 <motion.div initial={{ opacity: 0, y: 10 }} animate={{ opacity: 1, y: 0 }} transition={{
delay: 0.8 }} className="px-6 py-5 border rounded-xl bg-emerald-50/50 dark:bg-emerald-950/20 shadow-sm
flex gap-4 mt-6 print:bg-white print:border-gray-300 print:text-black">
508                 <Leaf className="h-6 w-6 text-emerald-600 shrink-0 mt-0.5" />
509                 <div className="space-y-2 text-foreground">
510                     <h3 className="text-sm font-bold text-emerald-800 dark:text-emerald-400 uppercase
tracking-widest">Agronomic Advisory</h3>
511                     <ul className="text-sm space-y-2 list-disc list-outside ml-4 text-emerald-900 dark:text-
emerald-100/90 font-medium">
512                         {diseaseData.disease !== "Healthy" && <li>Apply targeted bactericide or fungicide
mapping to {diseaseData.disease.replace(/_/g, " ")} immediately. Avoid systemic sprays before harvest.</
li>}
513                         {envData.uv_stress_level === "High" && <li>Deploy sunburn protection nets or kaolin
clay spray to reduce UV stress affecting surface Anthocyanin stability.</li>}
514                         <li>Harvest grading class should be adjusted to <strong>{nutritionData.quality_tier}</
strong> to match the inferred interior chemical composition.</li>
515                     </ul>
516                 </div>
517                 </motion.div>
518             </div>
519         </AnimatePresence>
520
521         {/* Actions */}

```

```
522         {isFinished && (  
523             <motion.div initial={{ opacity: 0 }} animate={{ opacity: 1 }} transition={{ delay: 1.2 }}  
               className="flex justify-end gap-4 mt-auto pt-6 no-print">  
524                 <Button variant="outline" size="lg" onClick={() => window.print()} className="font-semibold  
               shadow-sm"><Download className="h-4 w-4 mr-2" /> Download Report</Button>  
525                 <Button onClick={resetAll} size="lg" className="font-semibold shadow-sm"><RefreshCw  
               className="h-4 w-4 mr-2" /> Analyse Another Fruit</Button>  
526             </motion.div>  
527         ) }  
528     </div>  
529 </div>  
530 </div>  
531 </div>  
532 </div>  
533 )  
534 }
```

3.9 Agronomic Analytics Dashboard View (frontend/src/pages/Dashboard.jsx)

Relative Path: frontend/src/pages/Dashboard.jsx

```

1  import { useEffect, useState } from "react"
2  import { useAppStore } from "@store/useAppStore"
3  import { LineChart, Line, XAxis, YAxis, CartesianGrid, Tooltip, ResponsiveContainer } from "recharts"
4  import { getAnalysisHistory } from "@services/api"
5  import { Card, CardHeader, CardTitle, CardContent } from "@components/ui/card"
6  import { Badge } from "@components/ui/badge"
7  import 'leaflet/dist/leaflet.css';
8  import { MapContainer, TileLayer, CircleMarker, Popup } from 'react-leaflet';
9
10 export default function Dashboard() {
11   const { notifications } = useAppStore()
12   const [history, setHistory] = useState([])
13   const [loading, setLoading] = useState(true)
14
15   const activeAlerts = notifications.filter(n => !n.read).length
16
17   useEffect(() => {
18     async function fetchHistory() {
19       try {
20         const data = await getAnalysisHistory();
21         setHistory(data.reverse()); // Assume older to newer for chart
22       } catch (error) {
23         console.error(error);
24       } finally {
25         setLoading(false);
26       }
27     }
28     fetchHistory()
29   }, [])
30
31   const healthyFruits = history.filter((h) => (h.result?.disease || "") === "Healthy").length;
32   const healthRatio = history.length ? Math.round((healthyFruits / history.length) * 100) : 0;
33   const avgScore = history.length ? Math.round(history.reduce((acc, curr) => acc + (curr.result?.nutrition?.
     nutritional_score || 0), 0) / history.length) : 0;
34
35   // Format history for chart
36   const chartData = history.map(h => ({
37     date: new Date(h.created_at).toLocaleDateString(),
38     score: h.result?.nutrition?.nutritional_score || 0
39   })).reverse();
40
41   // Map Data
42   const scansWithLocation = history.filter(h => h.input_data?.latitude && h.input_data?.longitude);
43   const defaultCenter = [17.6599, 75.9064]; // Solapur
44   const mapCenter = scansWithLocation.length > 0
     ? [scansWithLocation[scansWithLocation.length - 1].input_data.latitude, scansWithLocation[
       scansWithLocation.length - 1].input_data.longitude]
     : defaultCenter;
45
46   const getMarkerColor = (severity) => {
47     switch (severity) {
48       case "Severe": return "#8B1A1A"; // Dark Red
49       case "Moderate": return "#D35400"; // Orange
50       case "Mild": return "#F1C40F"; // Yellow
51       case "None": default: return "#27AE60"; // Green
52     }
53   };
54
55   };
56
57   return (
58     <div className="container py-8 space-y-8 animate-in fade-in duration-500">
59       <div className="flex items-center justify-between">
60         <h1 className="text-3xl font-bold tracking-tight">Farmer Dashboard</h1>
61       </div>
62
63       <div className="grid grid-cols-1 md:grid-cols-4 gap-4">
64         <Card>
65           <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">Total
66             Analyses</CardTitle></CardHeader>
67           <CardContent><div className="text-3xl font-bold text-foreground">{loading ? "... " : history.length
68             }</div></CardContent>
69         </Card>
70         <Card>
71           <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">
72             Healthy Fruits %</CardTitle></CardHeader>
73           <CardContent><div className="text-3xl font-bold text-brand-success">{loading ? "... " : `${
74             healthRatio}%`}</div></CardContent>
75         </Card>
76         <Card>
77           <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">
78             Average Nutritional Score</CardTitle></CardHeader>
79           <CardContent><div className="text-3xl font-bold text-primary">{loading ? "... " : avgScore}</div></
80             CardContent>
81         </Card>
82       </div>
83     </div>
84   );

```

```

77       <CardHeader className="pb-2"><CardTitle className="text-sm font-medium text-muted-foreground">Active
Alerts</CardTitle></CardHeader>
78       <CardContent><div className="text-3xl font-bold text-destructive">{activeAlerts}</div></CardContent>
79     </Card>
80   </div>
81
82   <div className="grid grid-cols-1 lg:grid-cols-3 gap-8">
83     <Card className="lg:col-span-2 shadow-sm">
84       <CardHeader>
85         <CardTitle>Nutritional Score Over Time</CardTitle>
86       </CardHeader>
87       <CardContent>
88         <div className="h-[300px] w-full">
89           {loading ? (
90             <div className="w-full h-full flex items-center justify-center text-muted-foreground">Loading
chart...</div>
91           ) : (
92             <ResponsiveContainer width="100%" height="100%">
93               <LineChart data={chartData} margin={{ top: 10, right: 10, left: -20, bottom: 0 }}>
94                 <CartesianGrid strokeDasharray="3 3" vertical={false} stroke="hsl(var(--border))" />
95                 <XAxis dataKey="date" stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false}
axisLine={false} />
96                 <YAxis stroke="hsl(var(--muted-foreground))" fontSize={12} tickLine={false} axisLine={
false} domain={[0, 100]} />
97                 <Tooltip contentStyle={{ backgroundColor: 'hsl(var(--card))', borderColor: 'hsl(var(--
border))', color: 'hsl(var(--foreground))' }} />
98                 <Line type="monotone" dataKey="score" stroke="#8B1A1A" strokeWidth={3} dot={{ r: 4, fill:
'#8B1A1A' }} activeDot={{ r: 6 }} />
99               </LineChart>
100             </ResponsiveContainer>
101           )}
102         </div>
103       </CardContent>
104     </Card>
105
106     <Card className="shadow-sm">
107       <CardHeader>
108         <CardTitle>Farm Summary</CardTitle>
109       </CardHeader>
110       <CardContent className="space-y-4">
111         <div className="rounded-lg overflow-hidden border h-48 relative z-0">
112           {loading ? (
113             <div className="w-full h-full flex items-center justify-center text-muted-foreground bg-muted
/30">Loading Map...</div>
114           ) : (
115             <MapContainer center={mapCenter} zoom={15} style={{ height: "100%", width: "100%" }}
scrollWheelZoom={false}>
116               <TileLayer
117                 attribution='&copy; <a href="https://www.esri.com/">Esri</a>'
118                 url="https://server.arcgisonline.com/ArcGIS/rest/services/World_Imagery/MapServer/tile/{z
}/{y}/{x}"
119               />
120               {scansWithLocation.map((scan) => {
121                 const disease = scan.result?.disease || "Healthy";
122                 const severity = scan.result?.severity || "None";
123                 const color = getMarkerColor(severity);
124                 return (
125                   <CircleMarker
126                     key={scan.id}
127                     center={[scan.input_data.latitude, scan.input_data.longitude]}
128                     radius={8}
129                     pathOptions={{ color: 'white', weight: 2, fillColor: color, fillOpacity: 0.8 }}
130                   >
131                     <Popup>
132                       <div className="font-sans">
133                         <p className="font-bold text-sm m-0">{disease.replace(/_/g, " ")}</p>
134                         <p className="text-xs m-0 text-muted-foreground">{severity} Severity</p>
135                       </div>
136                     </Popup>
137                   </CircleMarker>
138                 );
139               })}
140             </MapContainer>
141           )}
142         </div>
143         <div className="space-y-2">
144           <div className="flex justify-between"><span className="text-muted-foreground text-sm">Location</
span><span className="font-medium text-sm">Solapur, Maharashtra</span></div>
145           <div className="flex justify-between"><span className="text-muted-foreground text-sm">Cultivar</
span><span className="font-medium text-sm">Bhagwa</span></div>
146           <div className="flex justify-between"><span className="text-muted-foreground text-sm">Total Area
</span><span className="font-medium text-sm">12 Acres</span></div>
147         </div>
148       </CardContent>
149     </Card>
150   </div>
151
152   <Card className="shadow-sm">
153     <CardHeader>
154       <CardTitle>Recent Analyses</CardTitle>

```

```

155     </CardHeader>
156     <CardContent>
157       <div className="overflow-x-auto">
158         <table className="w-full text-sm text-left">
159           <thead className="bg-muted/50 text-muted-foreground border-b">
160             <tr>
161               <th className="px-4 py-3 font-medium rounded-tl-md">Date</th>
162               <th className="px-4 py-3 font-medium">Disease Detected</th>
163               <th className="px-4 py-3 font-medium">Severity</th>
164               <th className="px-4 py-3 font-medium text-right">Score</th>
165               <th className="px-4 py-3 font-medium rounded-tr-md">Status</th>
166             </tr>
167           </thead>
168           <tbody>
169             {history.map((item) => {
170               const disease = item.result?.disease || "Unknown";
171               const severity = item.result?.severity || "None";
172               const score = item.result?.nutrition?.nutritional_score || 0;
173               const date = new Date(item.created_at).toLocaleDateString();
174
175               return (
176                 <tr key={item.id} className="border-b last:border-0 hover:bg-muted/30 transition-colors">
177                   <td className="px-4 py-3">{date}</td>
178                   <td className="px-4 py-3 font-medium text-foreground">{disease.replace(/_/g, " ")}</td>
179                   <td className="px-4 py-3">
180                     <Badge variant={severity === "None" ? "success" : severity === "Severe" ? "destructive"
181                       : "warning"}>
182                       {severity}
183                     </Badge>
184                   </td>
185                   <td className="px-4 py-3 text-right font-bold text-foreground">{score}</td>
186                   <td className="px-4 py-3 text-muted-foreground cursor-pointer hover:underline" onClick
187                     ={( () => window.location.href = `/scan/${item.id}`)}>View Analysis</td>
188                 </tr>
189               )
190             })}
191             {history.length === 0 && !loading && (
192               <tr>
193                 <td colSpan="5" className="text-center py-8 text-muted-foreground">No analyses found.</td>
194               </tr>
195             )}
196           </tbody>
197         </table>
198       </div>
199     </CardContent>
200   </Card>
201 </div>

```

3.10 Historical Diagnostic Reports View (frontend/src/pages/History.jsx)

Relative Path: frontend/src/pages/History.jsx

```

1  import { useEffect, useState } from "react"
2  import { getAnalysisHistory, deleteScan } from "@services/api"
3  import { Card, CardHeader, CardTitle, CardContent } from "@components/ui/card"
4  import { Badge } from "@components/ui/badge"
5  import { Button } from "@components/ui/button"
6  import { Trash2, Eye, Calendar, Leaf } from "lucide-react"
7  import { useNavigate } from "react-router-dom"
8  import { toast } from "sonner"
9
10 export default function History() {
11   const [history, setHistory] = useState([])
12   const [loading, setLoading] = useState(true)
13   const navigate = useNavigate()
14
15   useEffect(() => {
16     fetchHistory()
17   }, [])
18
19   const fetchHistory = async () => {
20     try {
21       setLoading(true)
22       const data = await getAnalysisHistory()
23       setHistory(data)
24     } catch (err) {
25       toast.error("Failed to fetch history")
26     } finally {
27       setLoading(false)
28     }
29   }
30
31   const handleDelete = async (id, e) => {
32     e.stopPropagation()
33     if (!confirm("Are you sure you want to delete this scan?")) return
34     try {
35       await deleteScan(id)
36       toast.success("Scan deleted")
37       fetchHistory()
38     } catch (err) {
39       toast.error("Failed to delete scan")
40     }
41   }
42
43   return (
44     <div className="container py-8 space-y-6 animate-in fade-in duration-500">
45       <div className="flex items-center justify-between">
46         <h1 className="text-3xl font-bold tracking-tight">Scan History</h1>
47       </div>
48
49       {loading ? (
50         <div className="text-muted-foreground text-center py-10">Loading history...</div>
51       ) : history.length === 0 ? (
52         <div className="text-center py-20 bg-muted/20 border rounded-lg">
53           <Leaf className="mx-auto h-12 w-12 text-muted-foreground opacity-50 mb-4" />
54           <h2 className="text-xl font-medium">No Scans Found</h2>
55           <p className="text-muted-foreground mt-2 mb-4">You haven't performed any fruit analyses yet.</p>
56           <Button onClick={() => navigate('/analyse')}>Run a Scan</Button>
57         </div>
58       ) : (
59         <div className="grid grid-cols-1 md:grid-cols-2 lg:grid-cols-3 gap-6">
60           {history.map((scan) => {
61             const disease = scan.result?.disease || "Unknown"
62             const severity = scan.result?.severity || "None"
63             const score = scan.result?.nutrition?.nutritional_score || 0
64             const date = new Date(scan.created_at).toLocaleDateString(undefined, {
65               year: 'numeric', month: 'short', day: 'numeric'
66             })
67
68             return (
69               <Card key={scan.id} className="cursor-pointer hover:shadow-md transition-shadow group overflow-hidden"
70                 onClick={() => navigate(`/scan/${scan.id}`)}>
71                 <div className="h-48 w-full bg-muted border-b relative">
72                   <img src={scan.media_url} alt={disease} className="w-full h-full object-cover" />
73                   <div className="absolute inset-0 bg-black/40 opacity-0 group-hover:opacity-100 transition-opacity flex items-center justify-center">
74                     <Button variant="secondary" size="sm" className="pointer-events-none"><Eye className="h-4 w-4 mr-2"/>View Details</Button>
75                   </div>
76                 </div>
77                 <CardContent className="p-4 space-y-4">
78                   <div className="flex justify-between items-start">
79                     <div>
80                       <h3 className="font-bold text-lg text-foreground line-clamp-1">{disease.replace(/_/g, "

```

```
81         <Calendar className="h-3 w-3 mr-1" /> {date}
82     </div>
83 </div>
84     <Badge variant={severity === "None" ? "success" : severity === "Severe" ? "destructive" :
"warning"}>
85         {severity}
86     </Badge>
87 </div>
88
89     <div className="flex items-center justify-between pt-2 border-t text-sm">
90         <div>
91             <span className="text-muted-foreground">Score: </span>
92             <span className="font-bold">{score} / 100</span>
93         </div>
94         <Button variant="ghost" size="icon" className="h-8 w-8 text-muted-foreground hover:text-
destructive z-10" onClick={ (e) => handleDelete(scan.id, e)}>
95             <Trash2 className="h-4 w-4" />
96         </Button>
97     </div>
98 </CardContent>
99 </Card>
100 )
101 }}
102 </div>
103 )
104 </div>
105 )
106 }
```